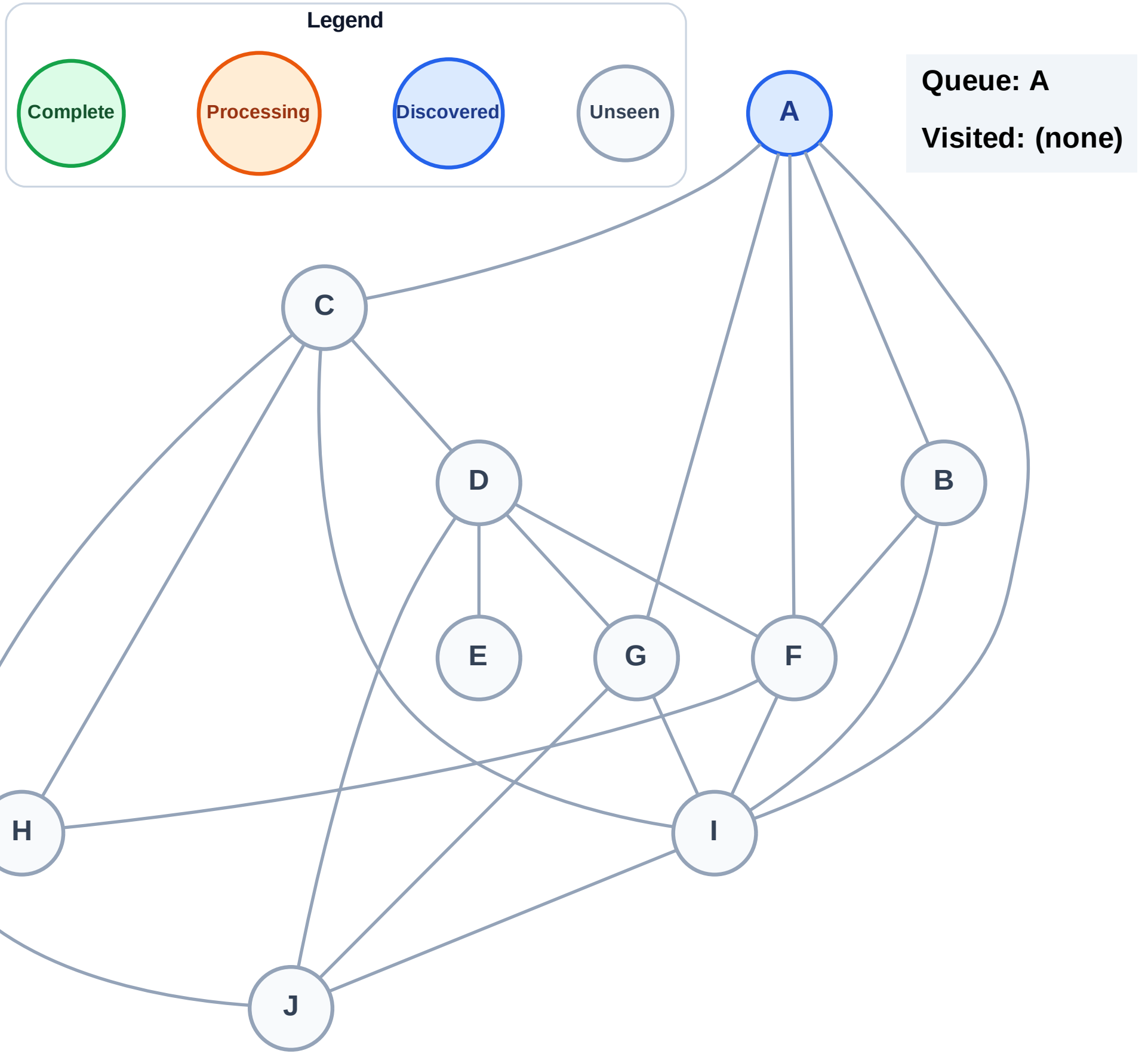
BFS Traversal

Step 1: Start at A

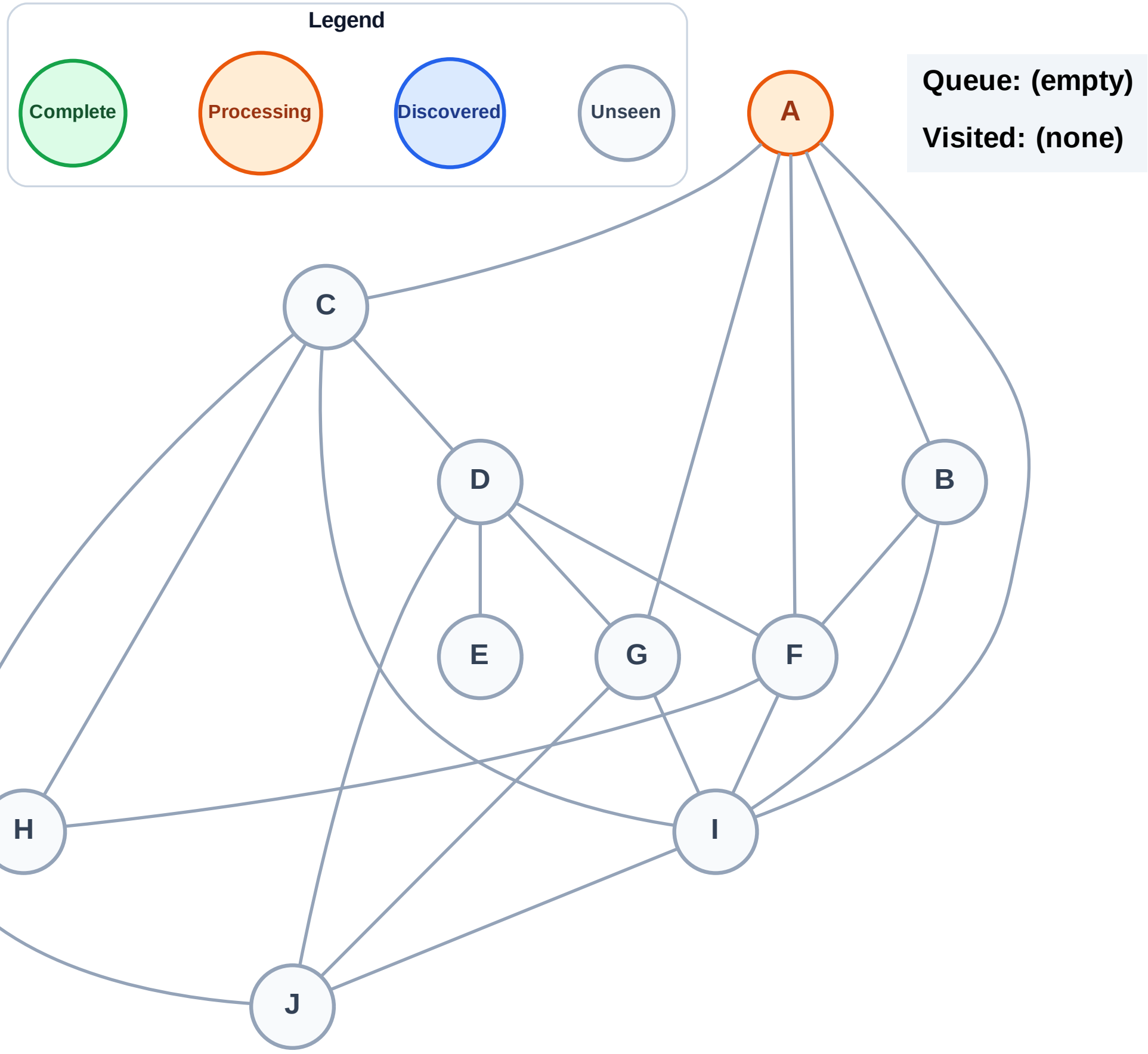


Queue: A

Visited: (none)

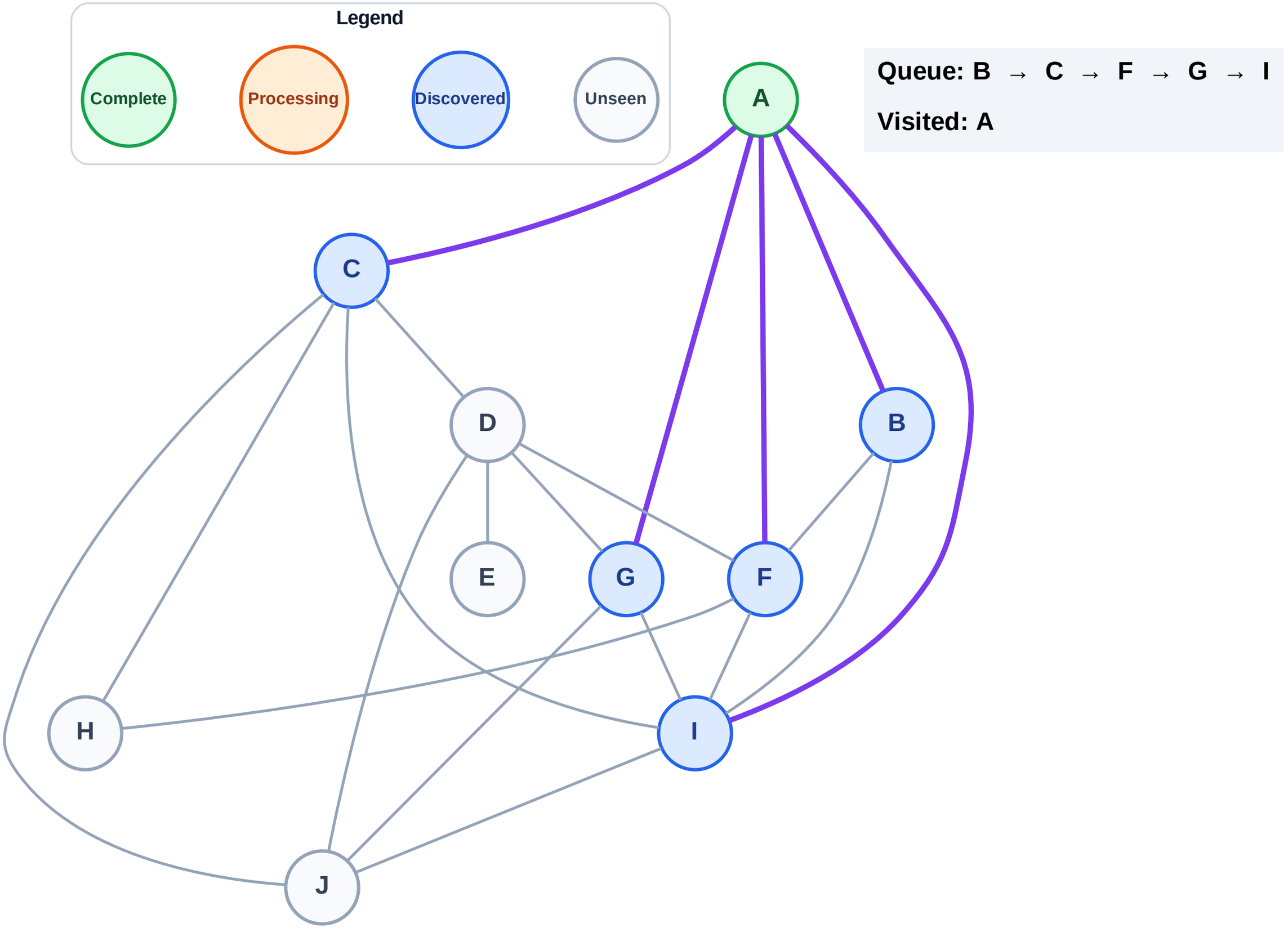
BFS Traversal

Step 2: Process node A



BFS Traversal

Step 3: Discover B, C, F, G, I from A

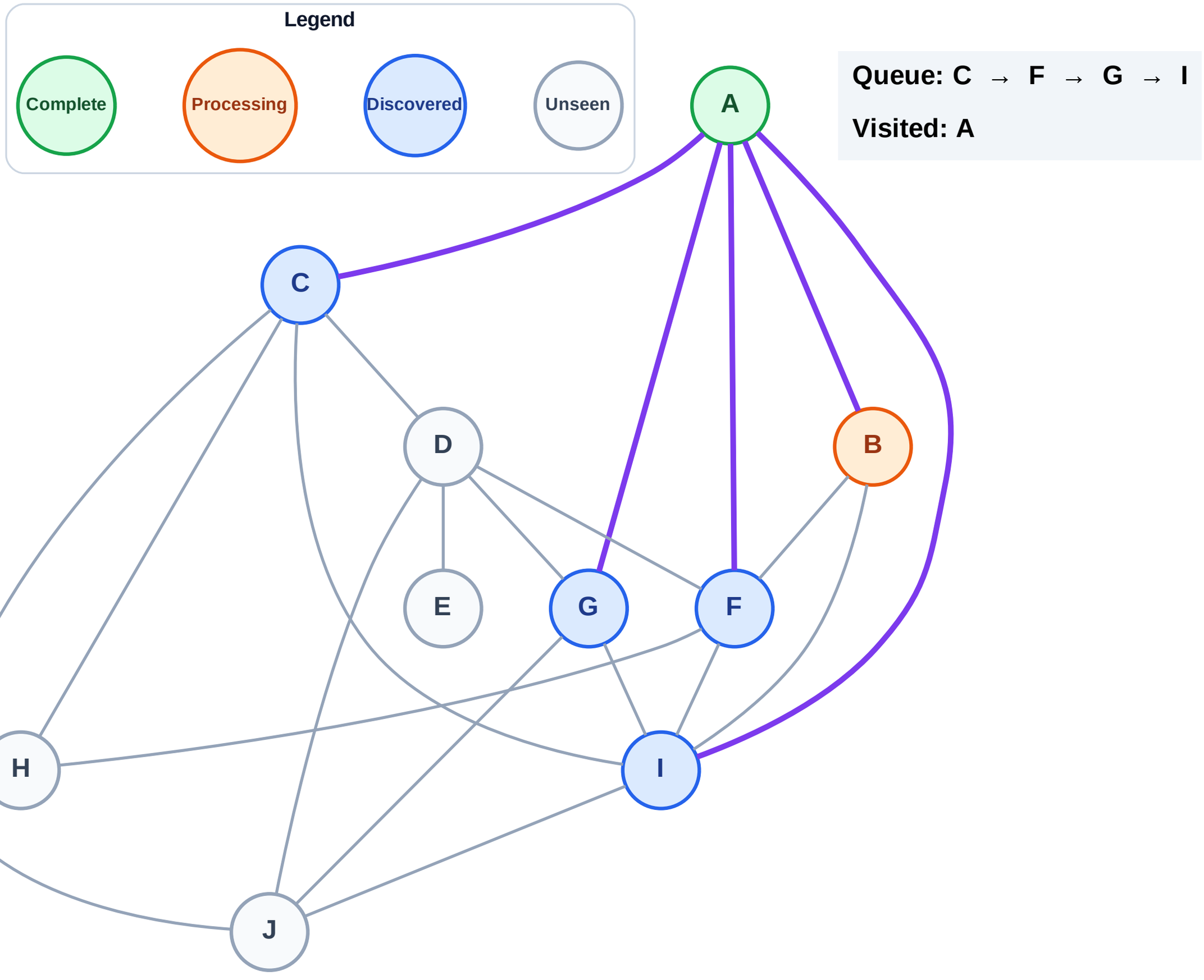


Queue: B → C → F → G → I

Visited: A

BFS Traversal

Step 4: Process node B

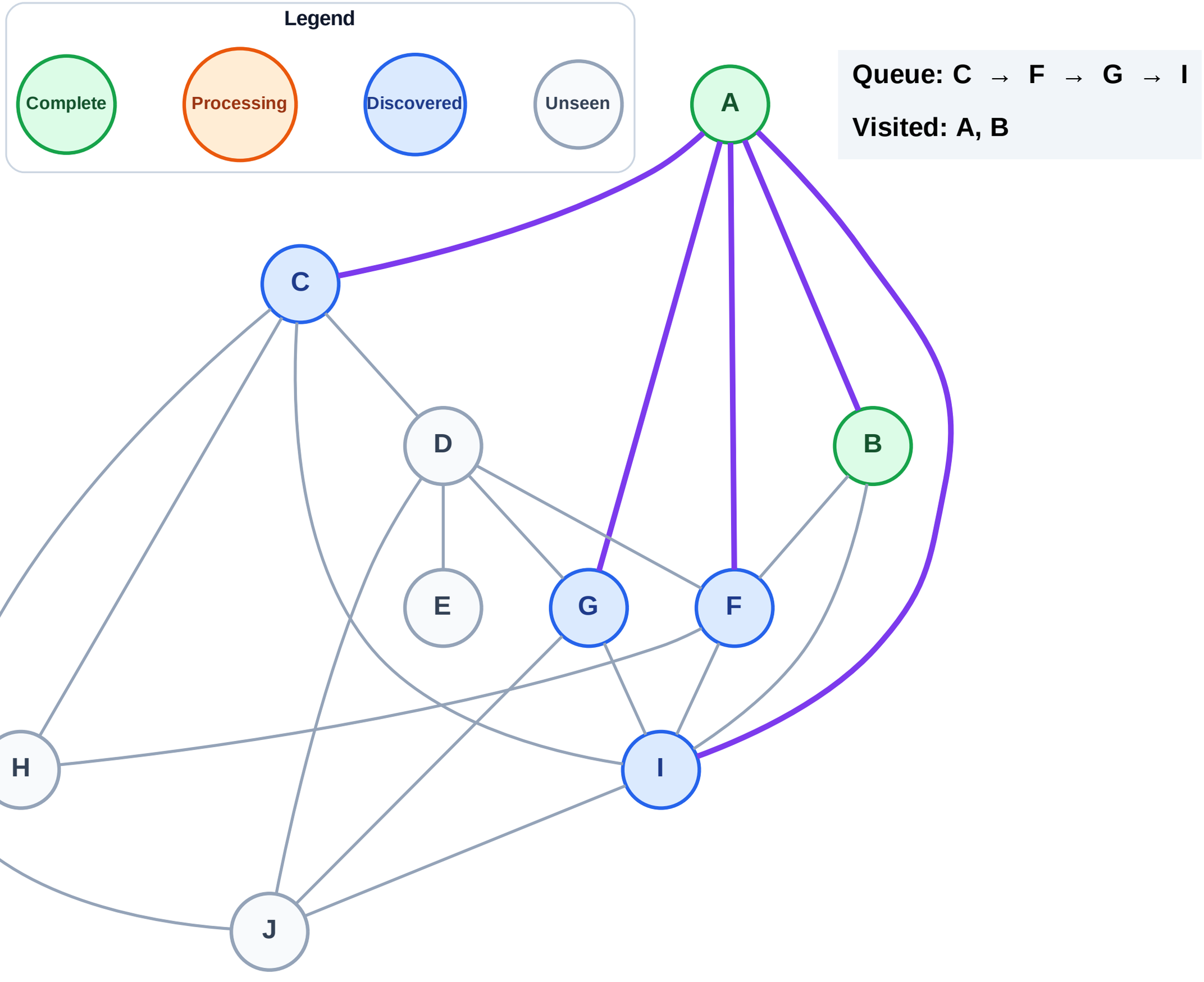


Queue: C → F → G → I

Visited: A

BFS Traversal

Step 5: No new neighbors from B

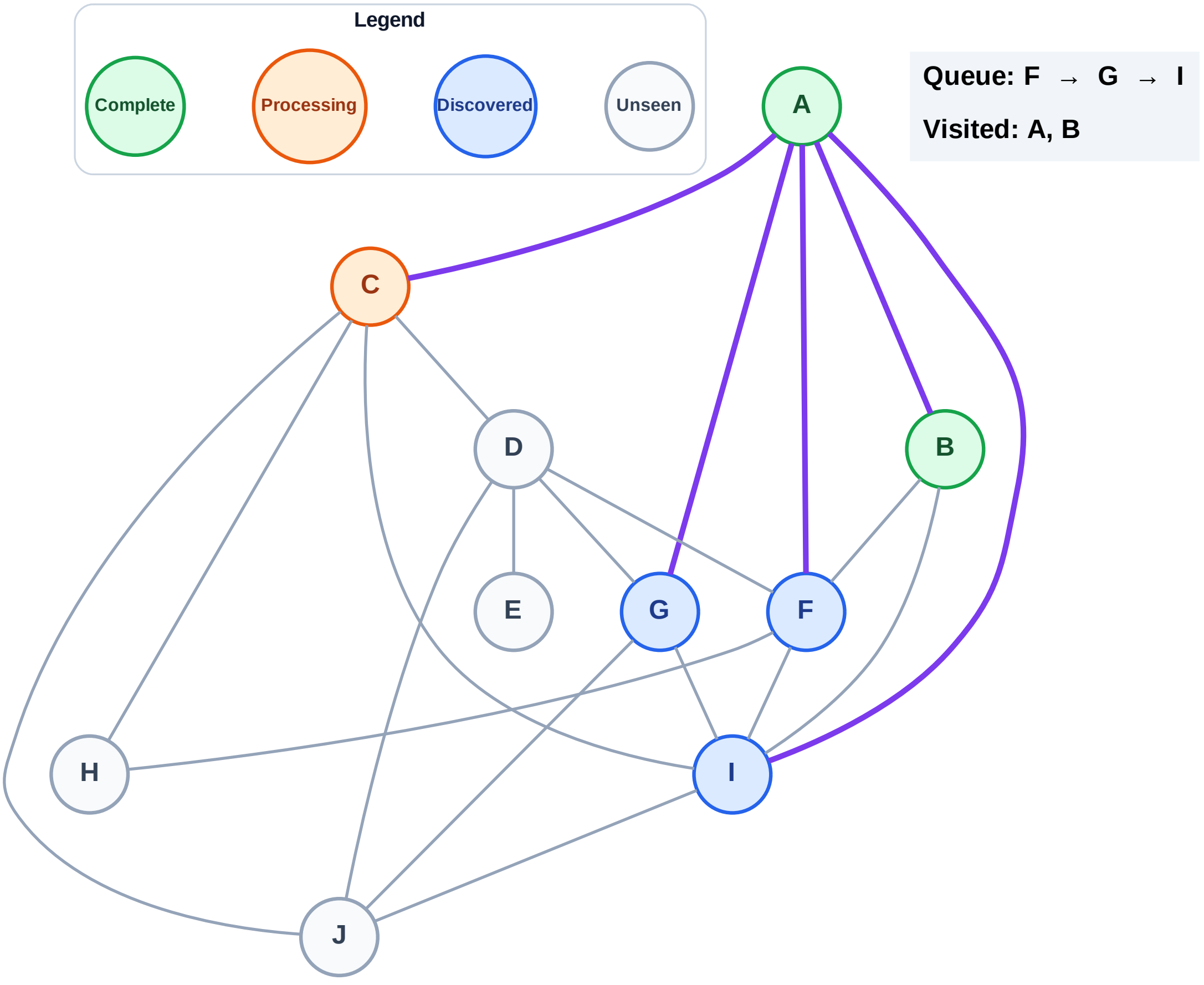


Queue: C → F → G → I

Visited: A, B

BFS Traversal

Step 6: Process node C

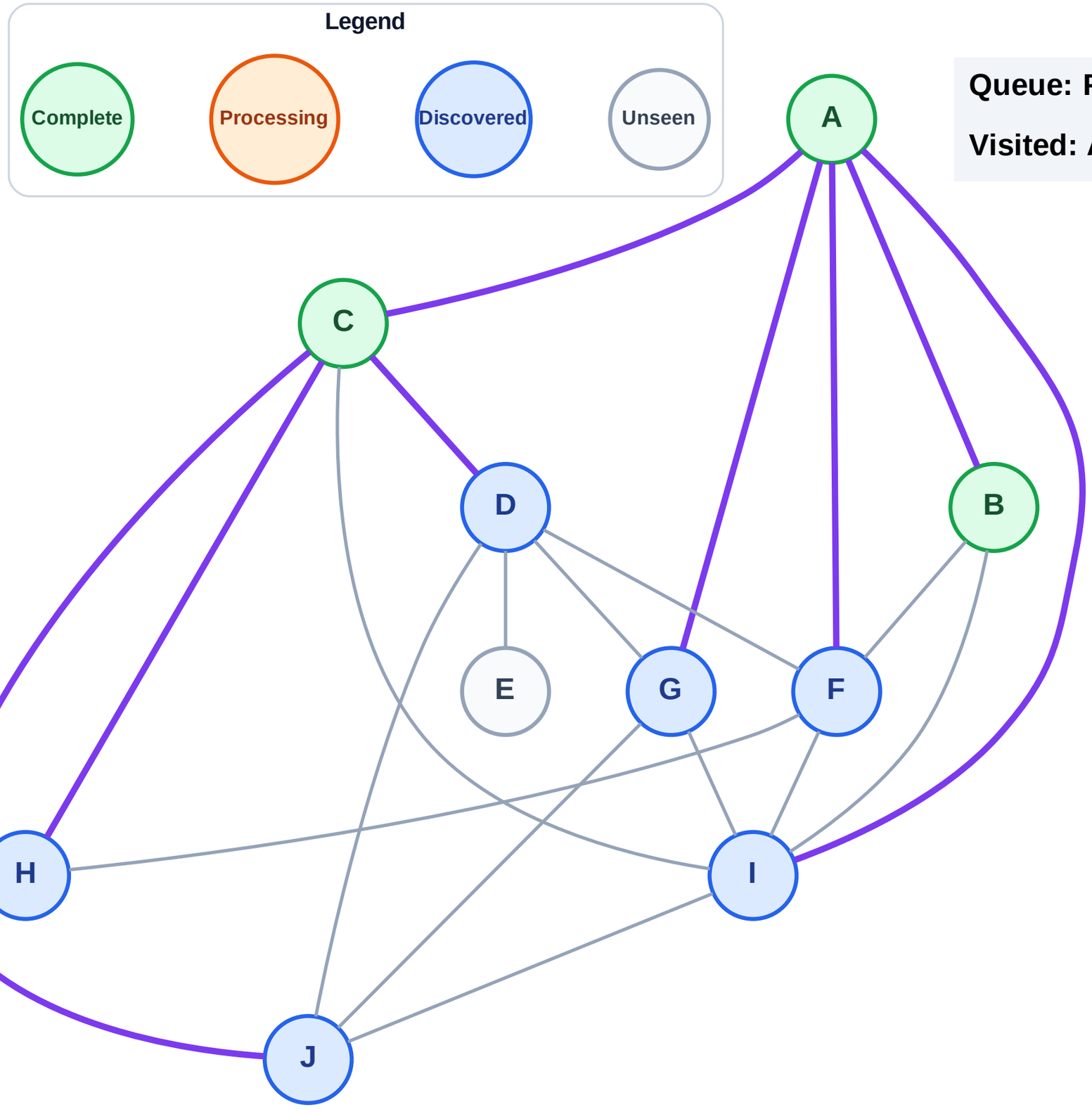


Queue: F → G → I

Visited: A, B

BFS Traversal

Step 7: Discover D, H, J from C

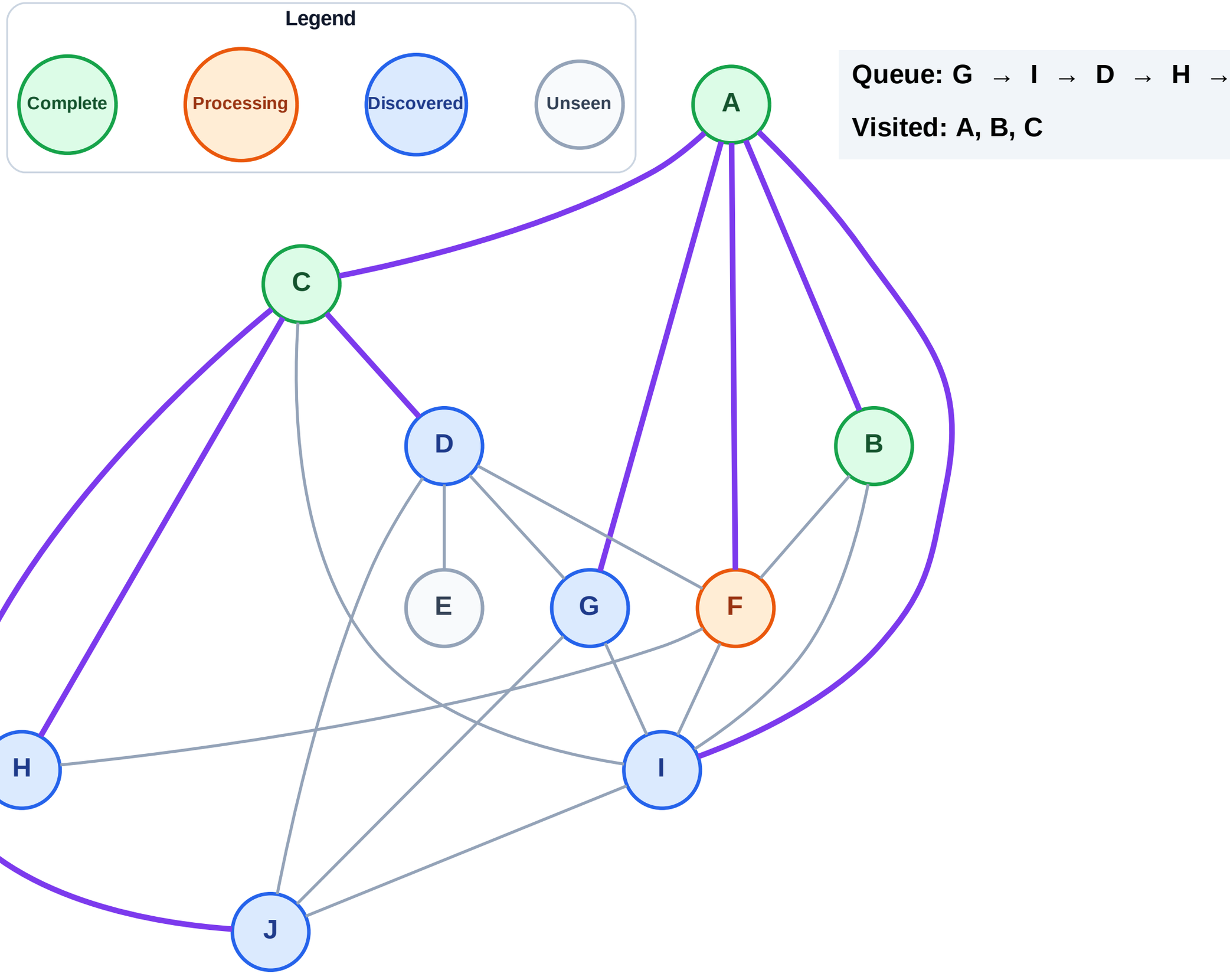


Queue: F → G → I → D → H → J

Visited: A, B, C

BFS Traversal

Step 8: Process node F

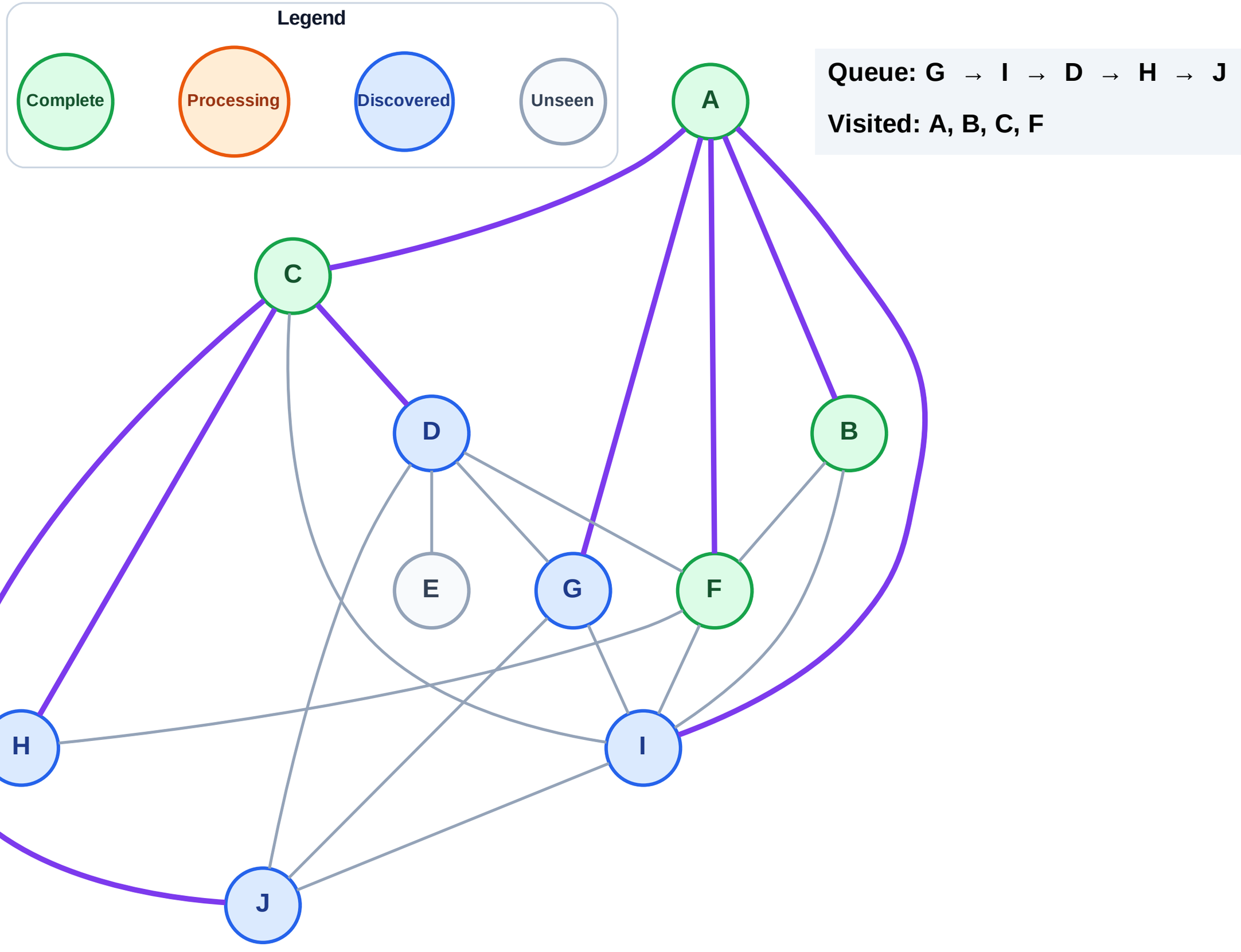


Queue: G → I → D → H → J

Visited: A, B, C

BFS Traversal

Step 9: No new neighbors from F

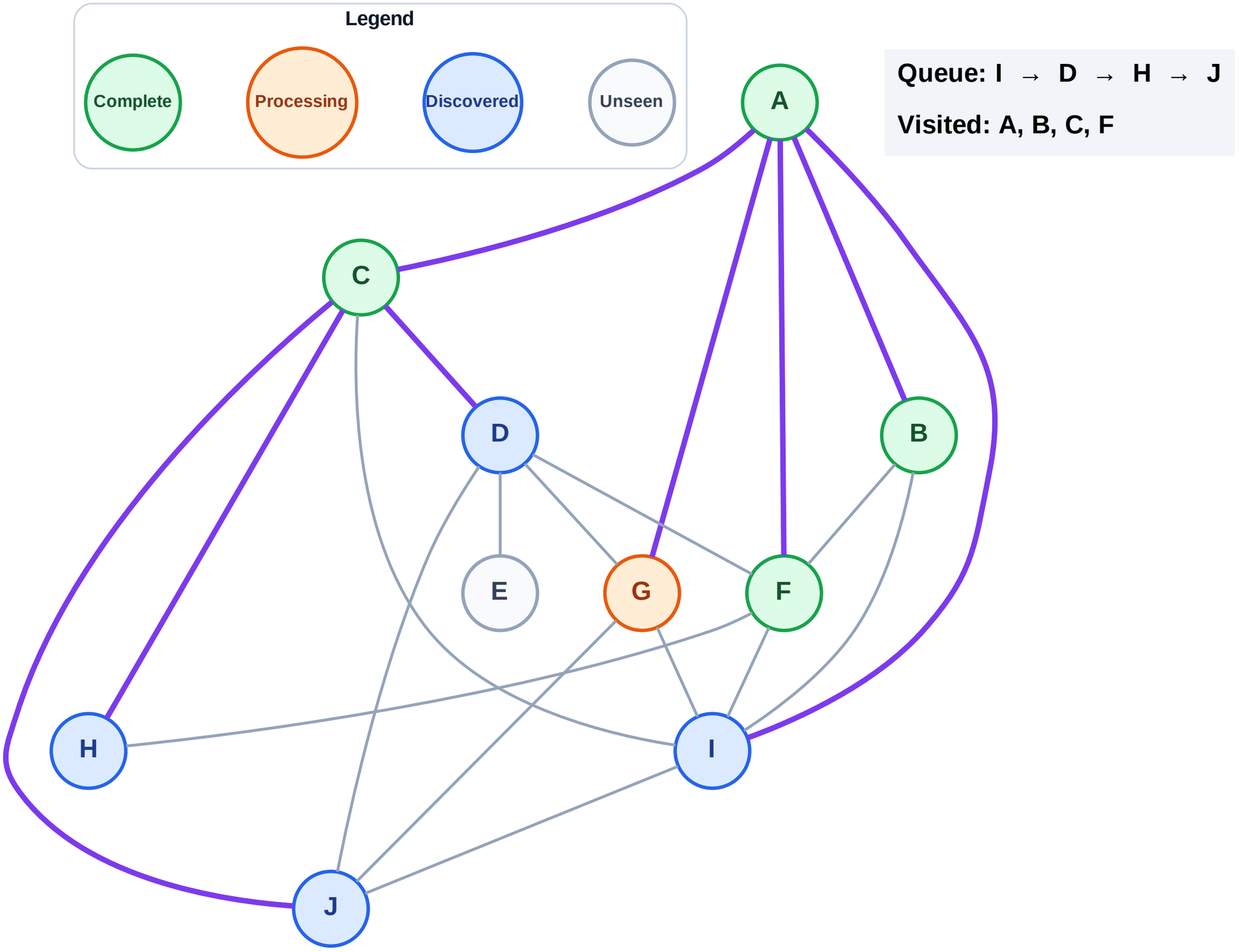


Queue: G → I → D → H → J

Visited: A, B, C, F

BFS Traversal

Step 10: Process node G

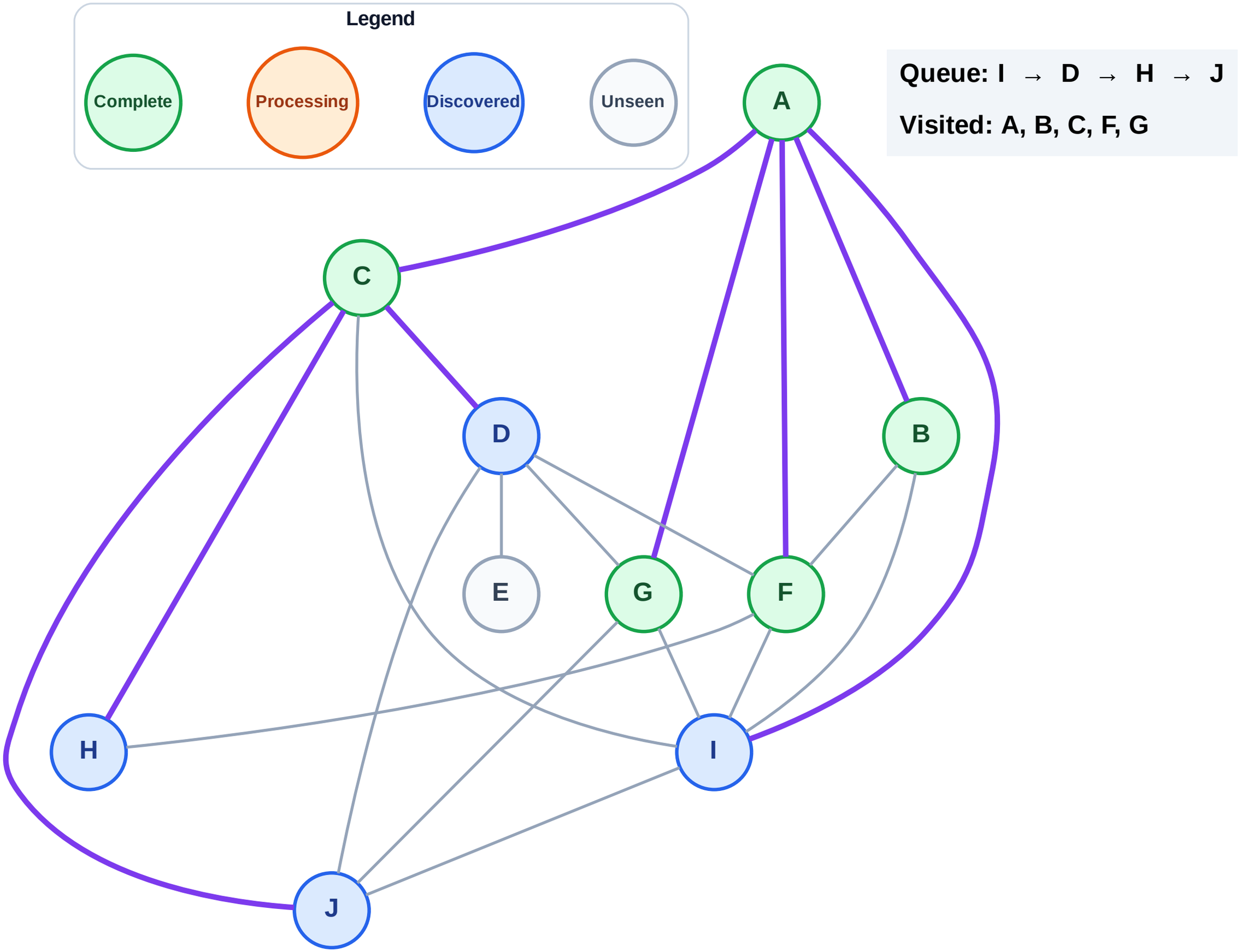


Queue: I → D → H → J

Visited: A, B, C, F

BFS Traversal

Step 11: No new neighbors from G

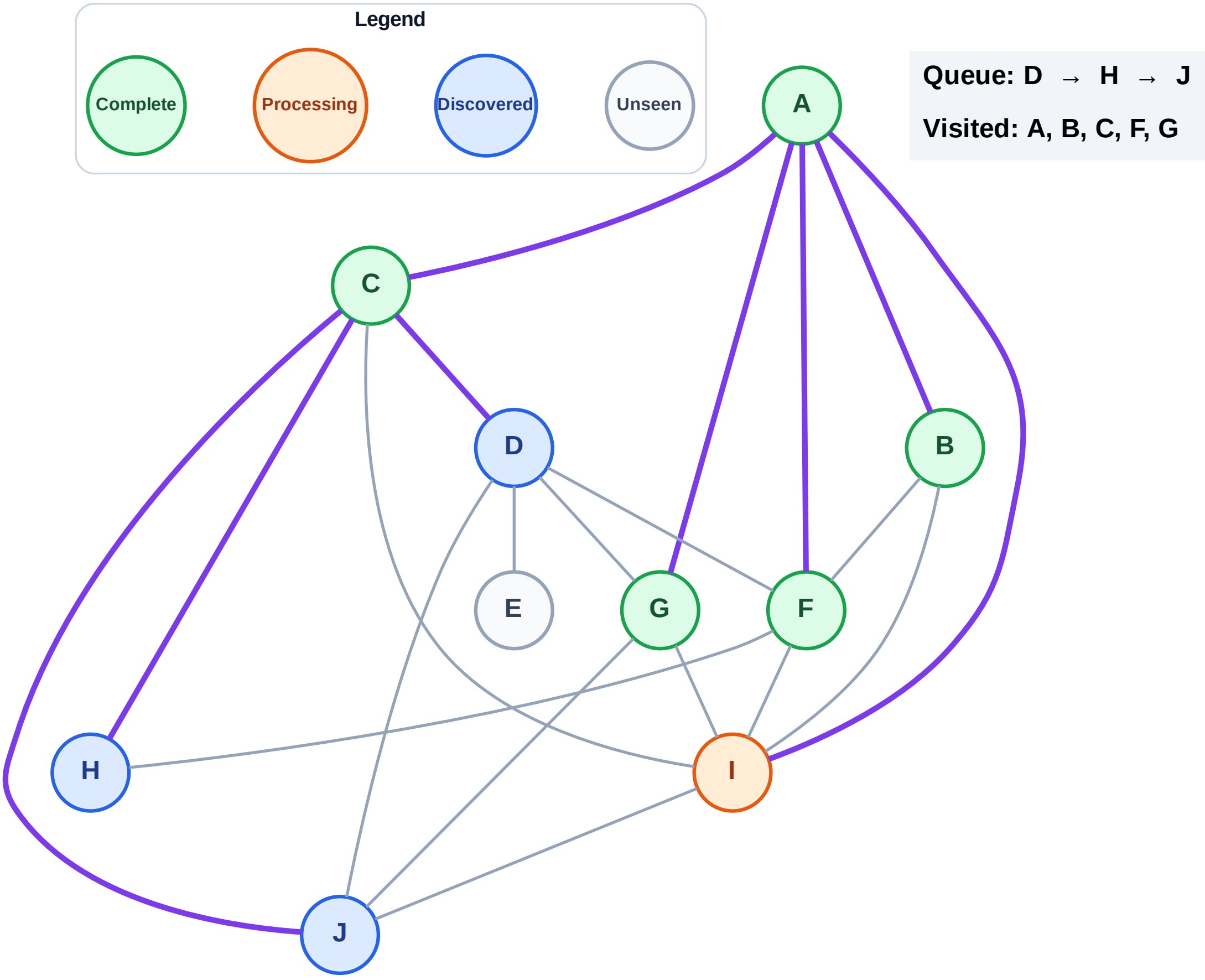


Queue: I → D → H → J

Visited: A, B, C, F, G

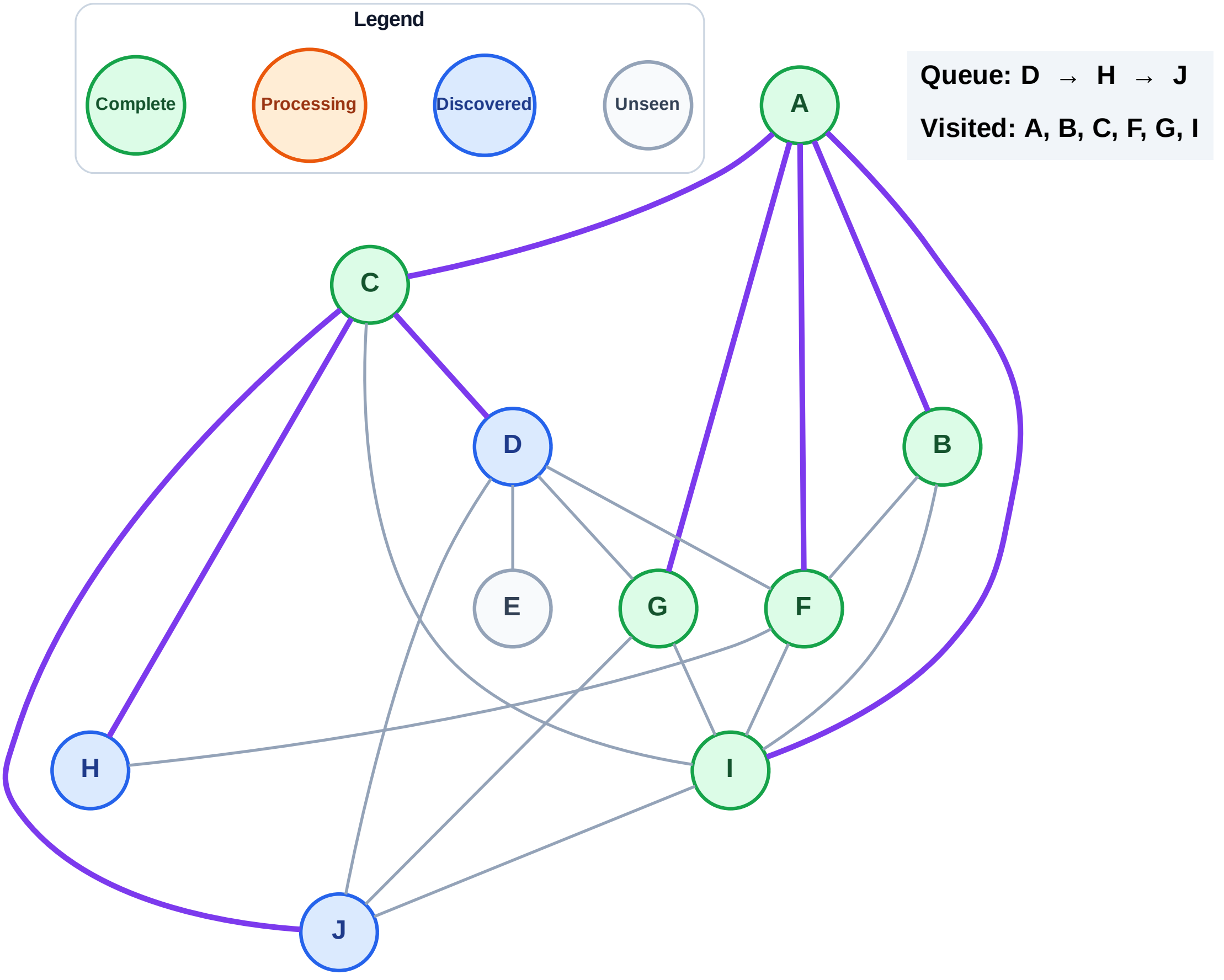
BFS Traversal

Step 12: Process node I



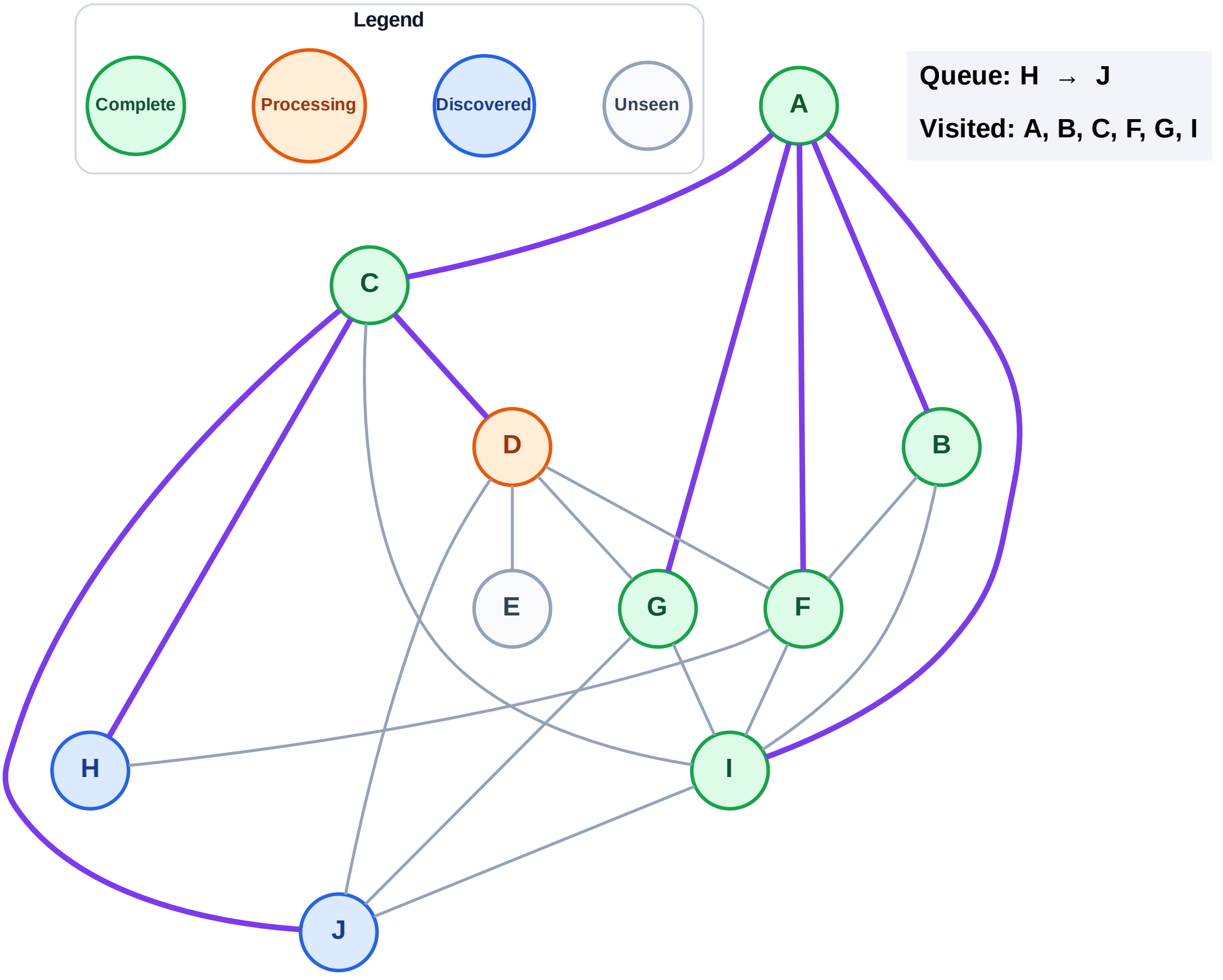
BFS Traversal

Step 13: No new neighbors from I



BFS Traversal

Step 14: Process node D

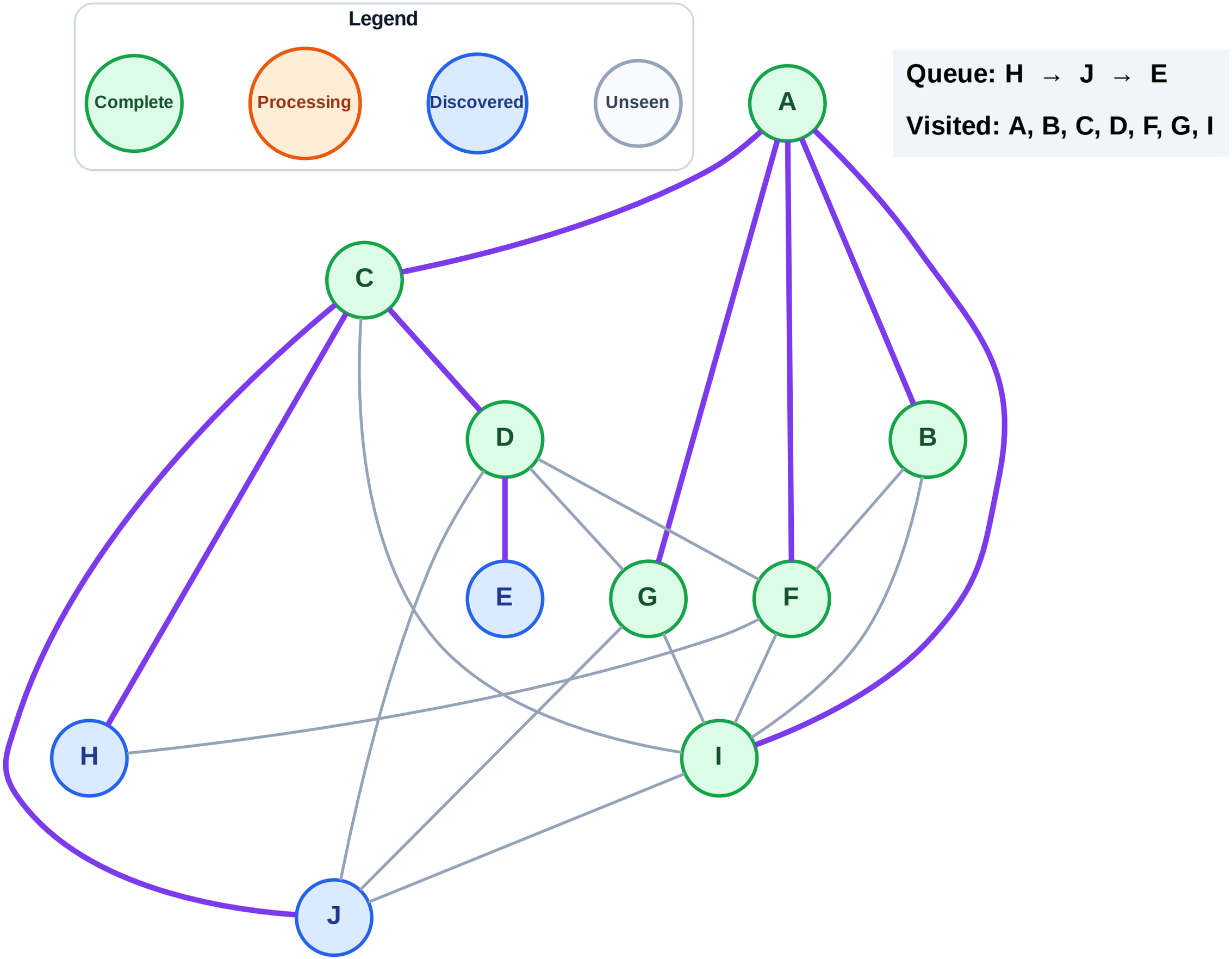


Queue: H → J

Visited: A, B, C, F, G, I

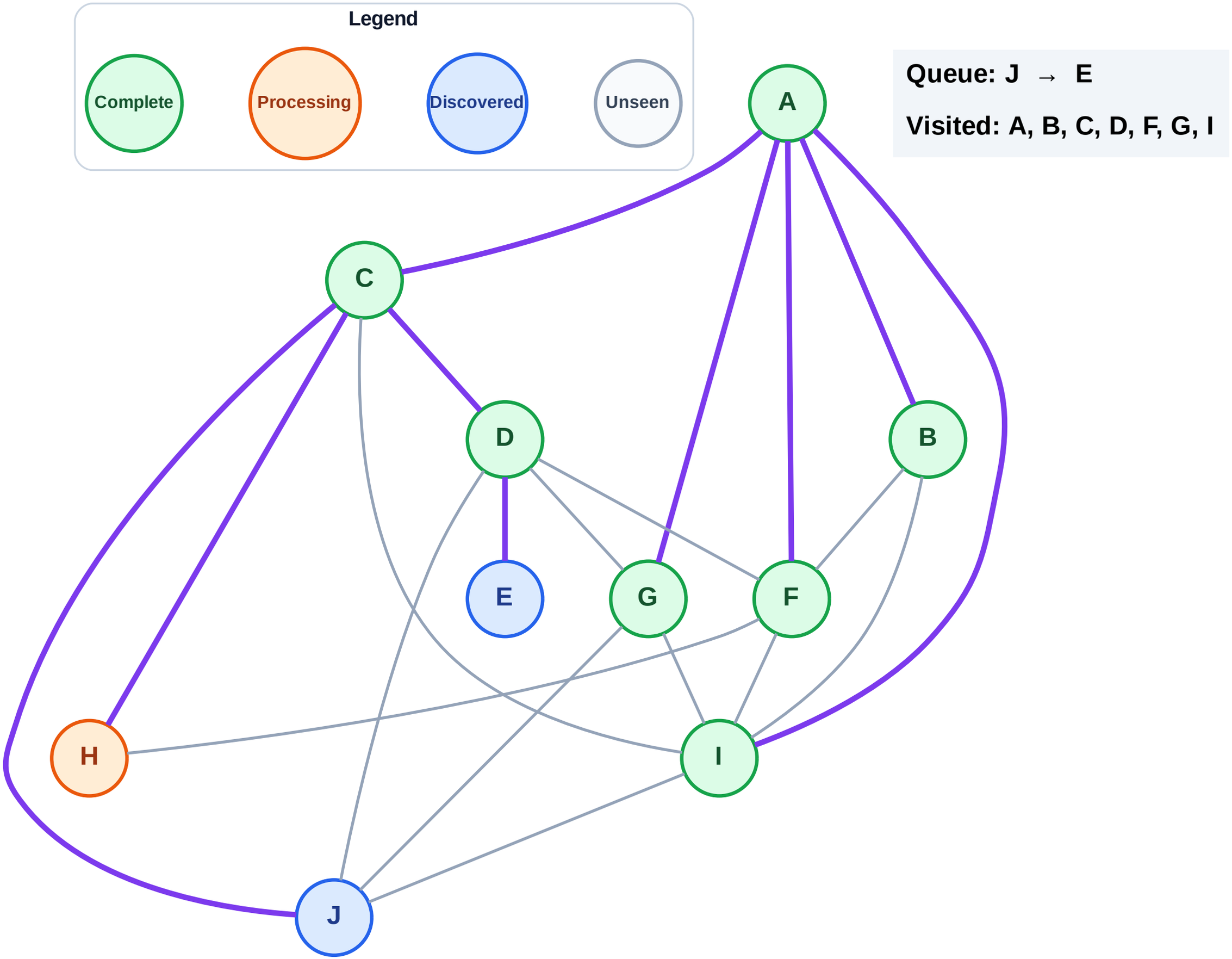
BFS Traversal

Step 15: Discover E from D



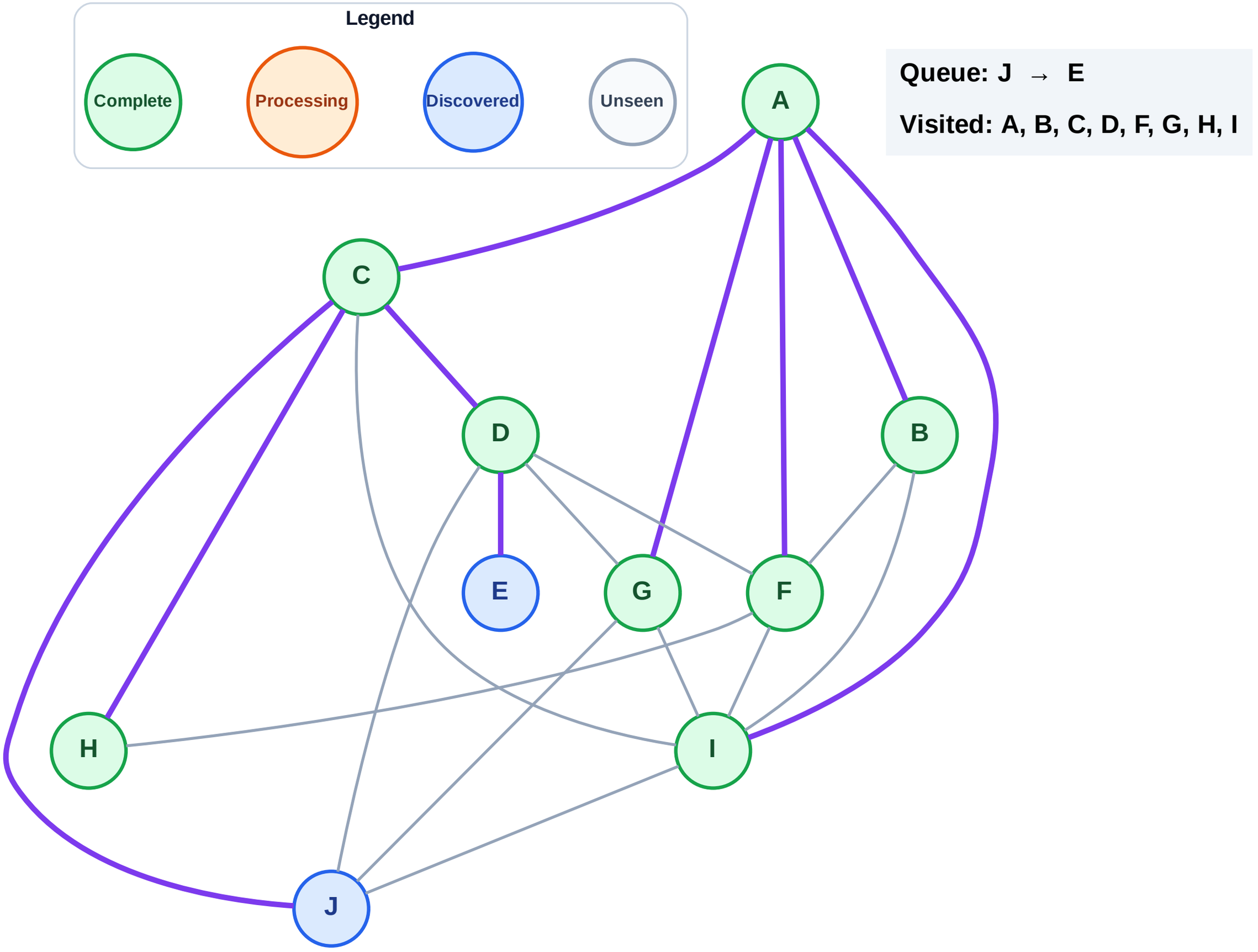
BFS Traversal

Step 16: Process node H



BFS Traversal

Step 17: No new neighbors from H

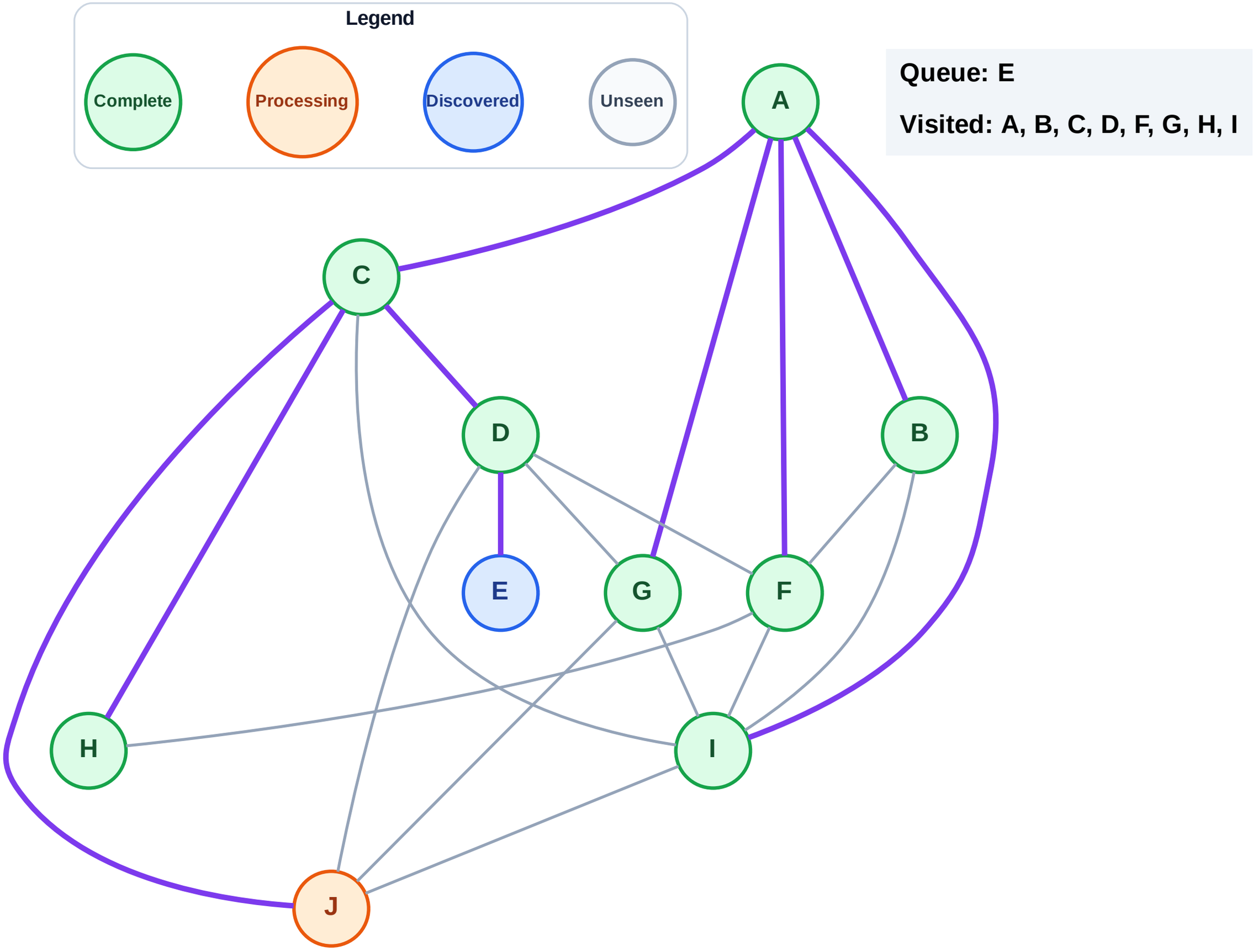


Queue: J → E

Visited: A, B, C, D, F, G, H, I

BFS Traversal

Step 18: Process node J

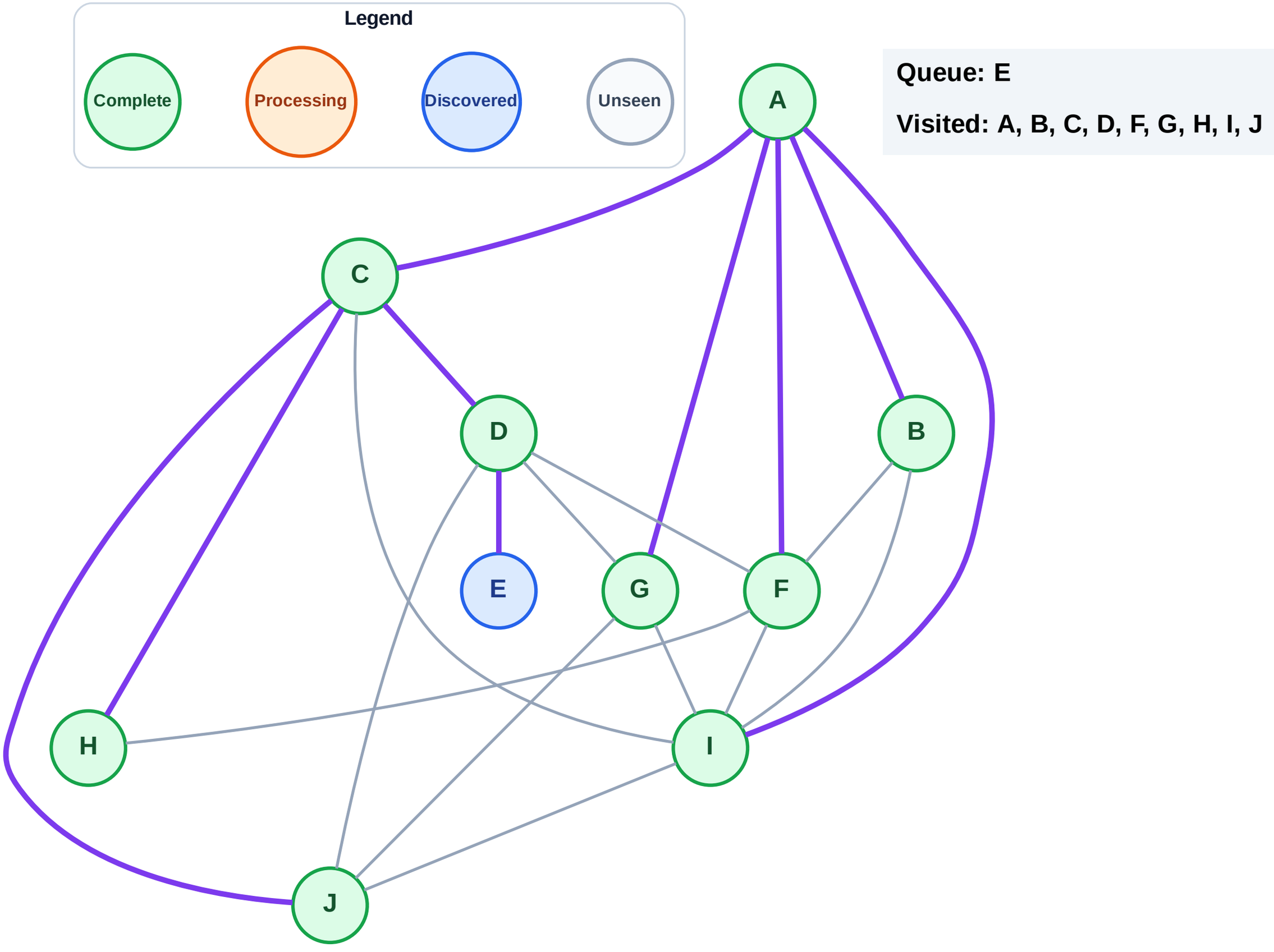


Queue: E

Visited: A, B, C, D, F, G, H, I

BFS Traversal

Step 19: No new neighbors from J

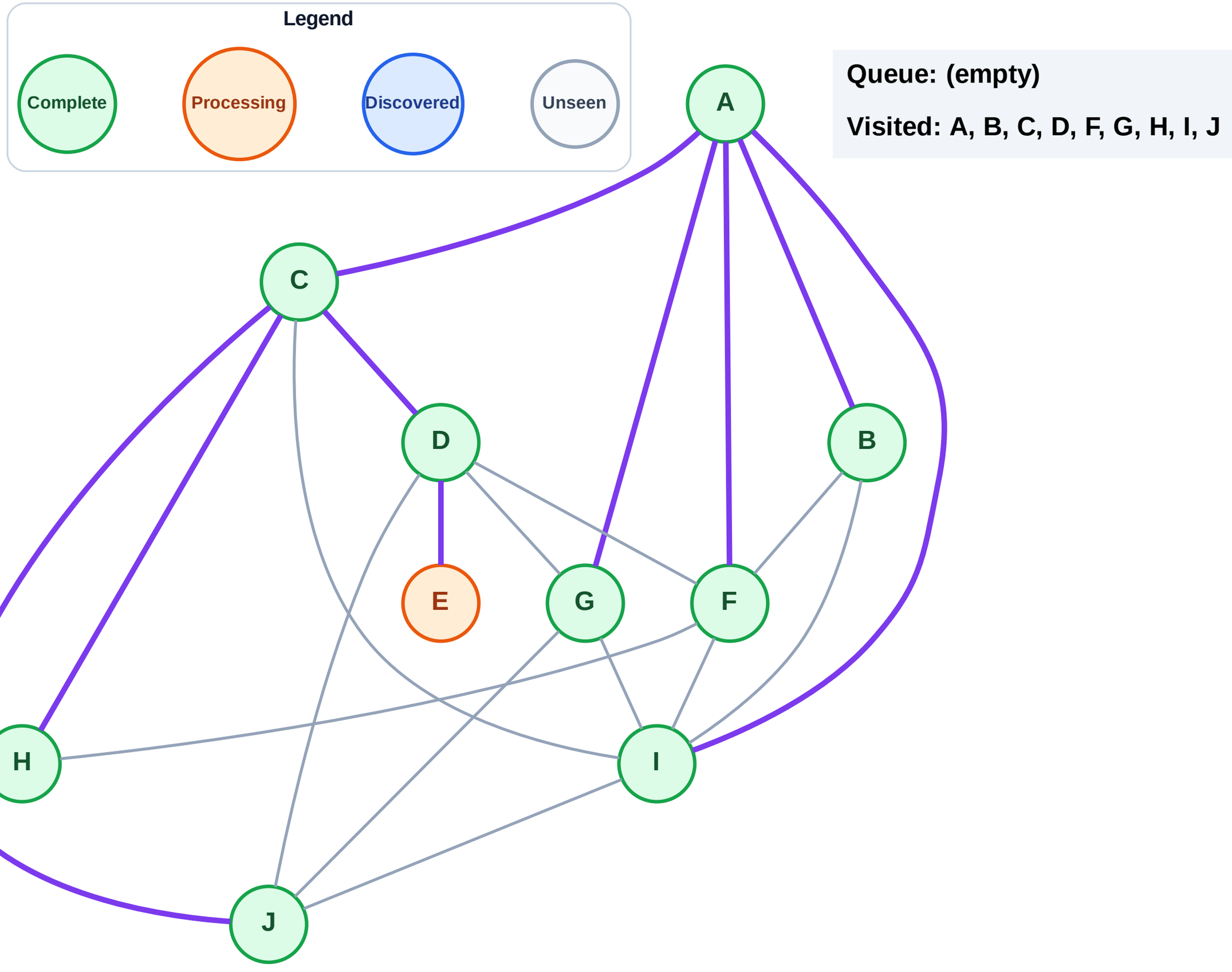


Queue: E

Visited: A, B, C, D, F, G, H, I, J

BFS Traversal

Step 20: Process node E

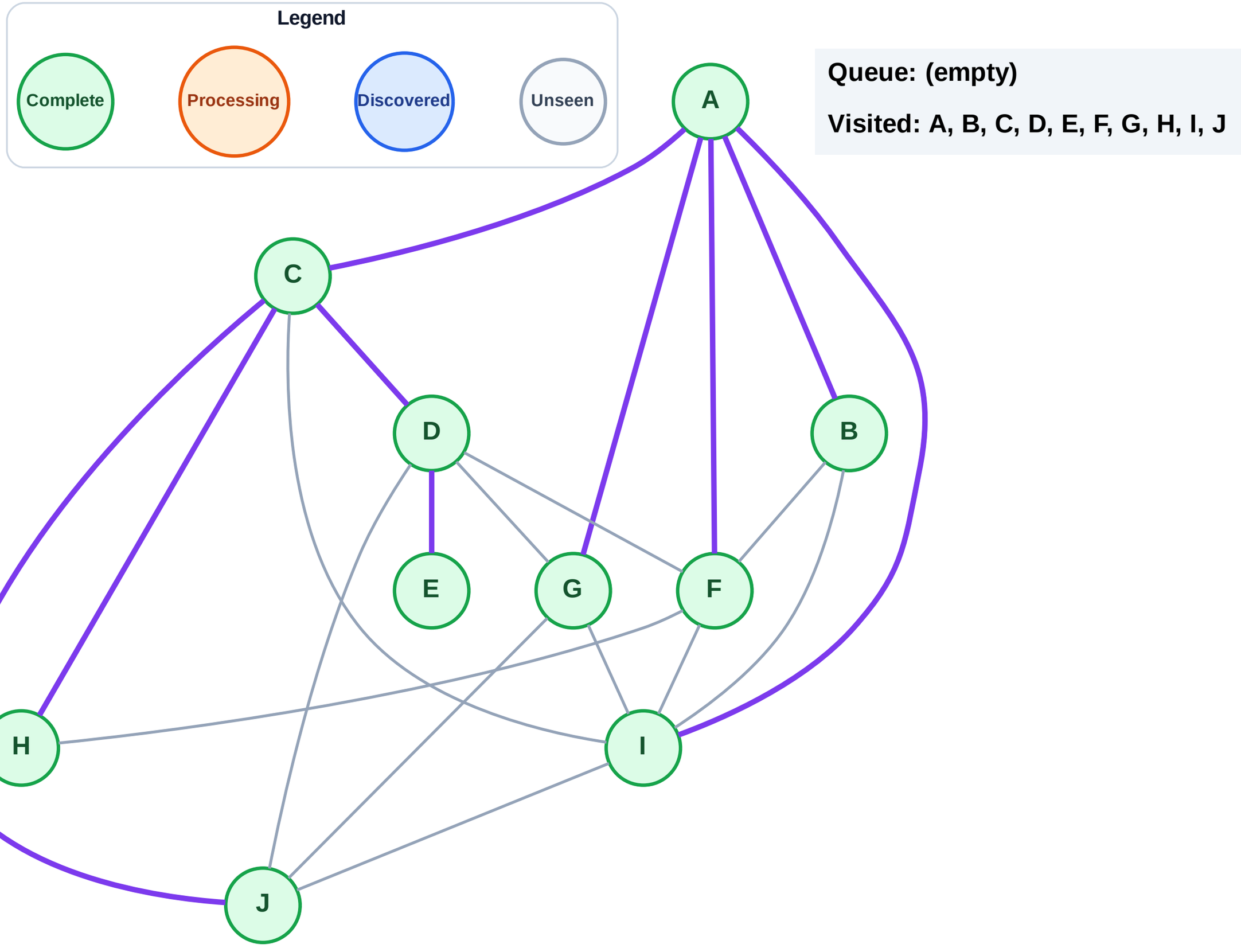


Queue: (empty)

Visited: A, B, C, D, F, G, H, I, J

BFS Traversal

Step 21: No new neighbors from E

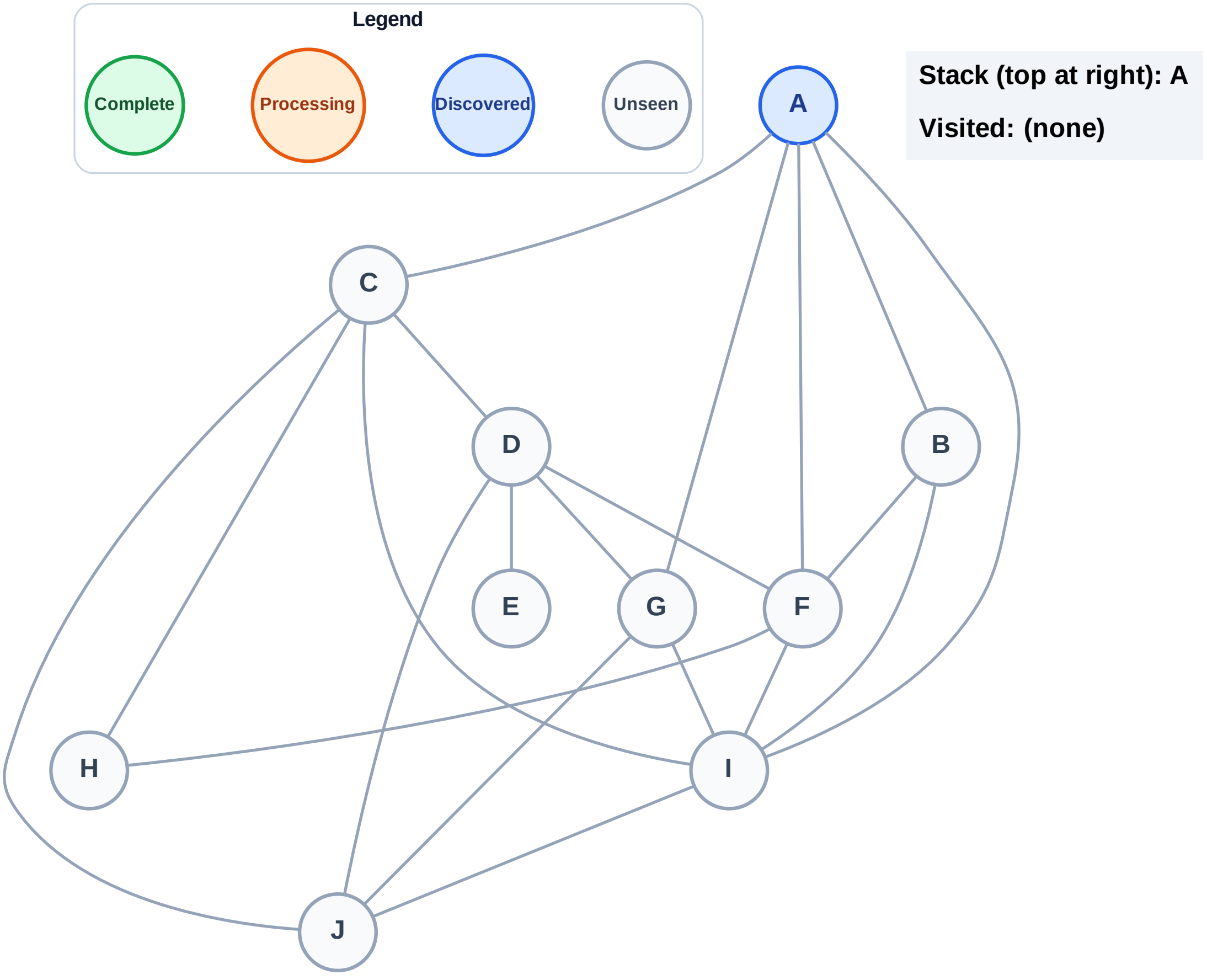


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

DFS Traversal

Step 1: Start at A

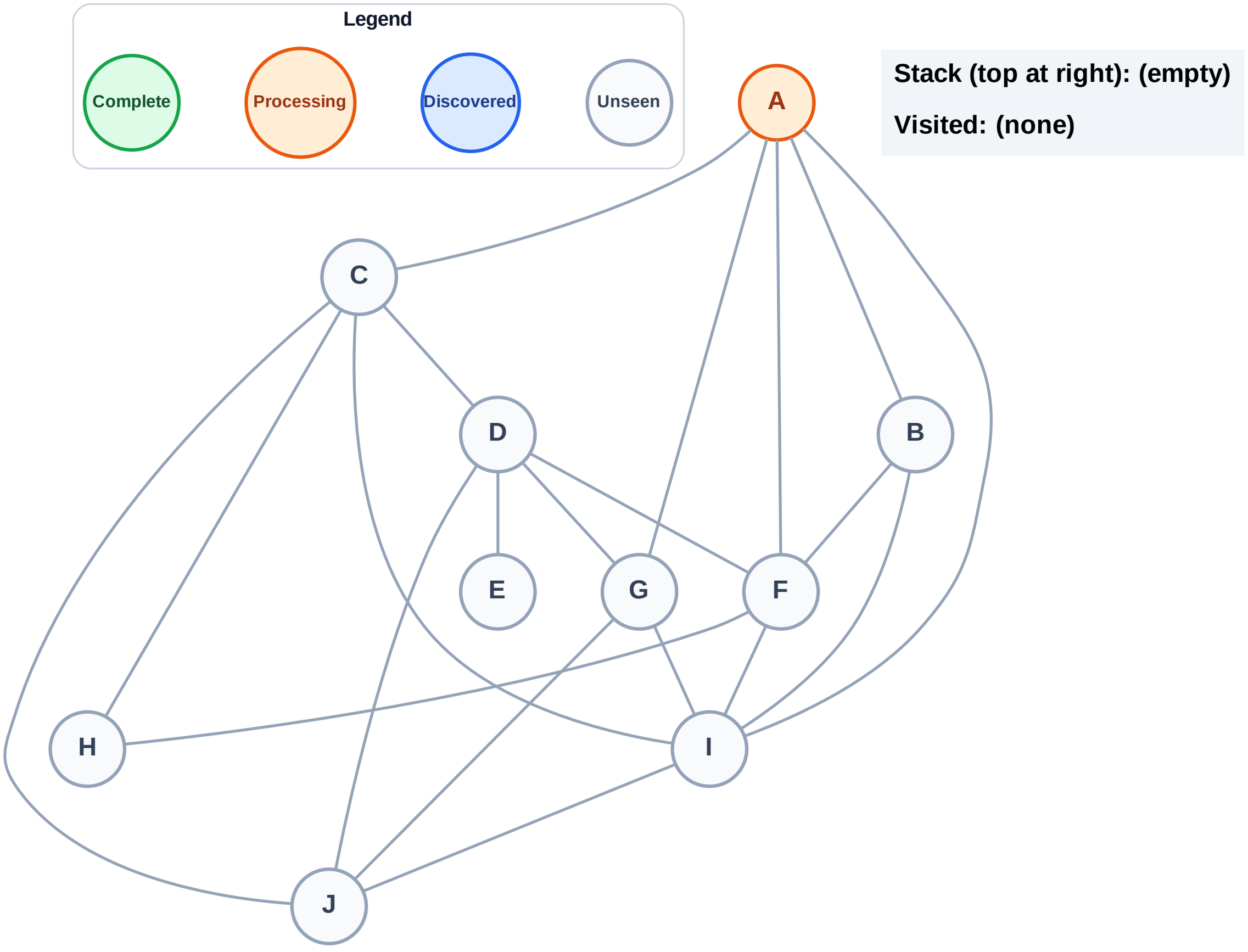


Stack (top at right): A

Visited: (none)

DFS Traversal

Step 2: Process node A

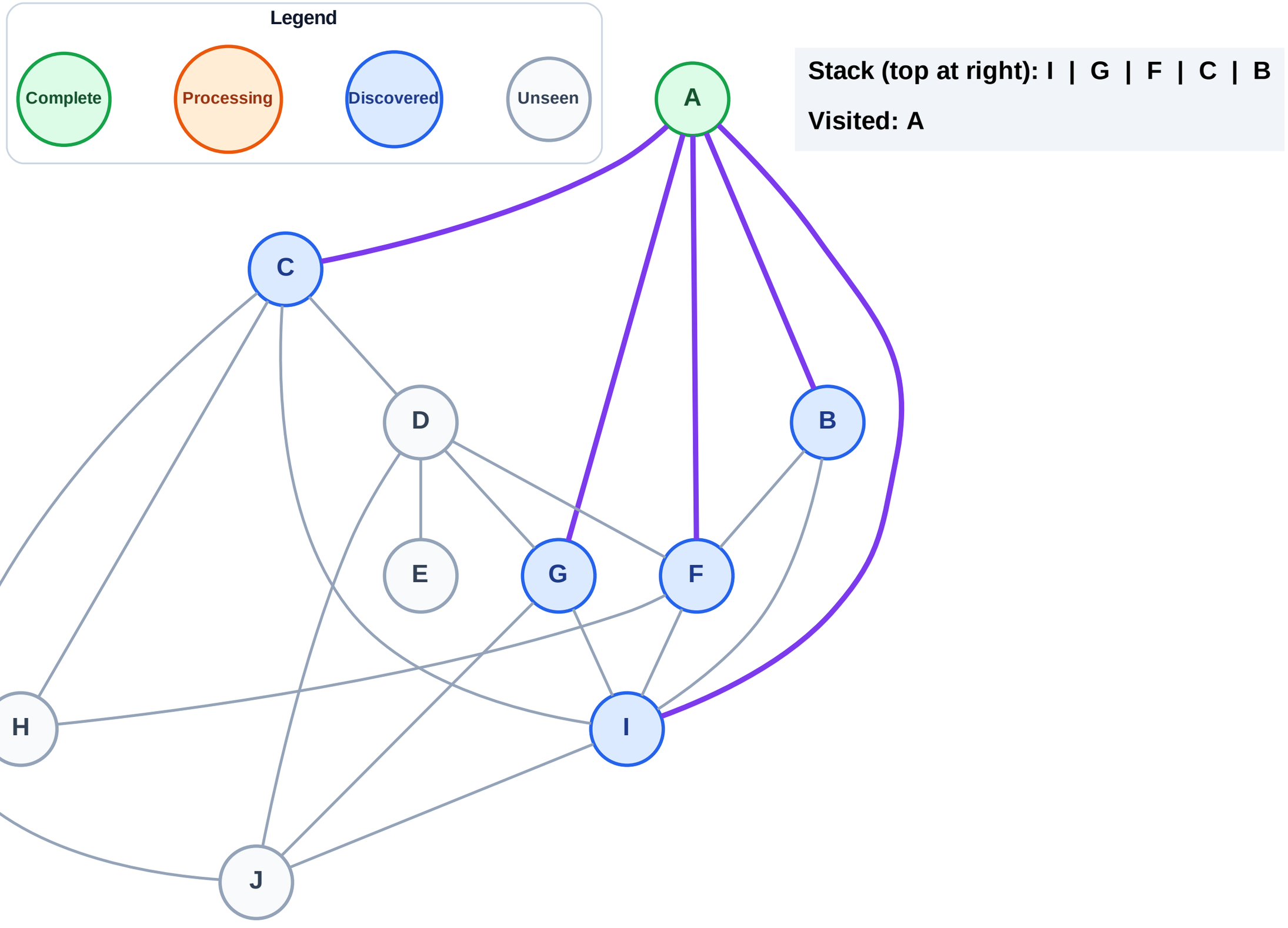


Stack (top at right): (empty)

Visited: (none)

DFS Traversal

Step 3: Discover I, G, F, C, B from A

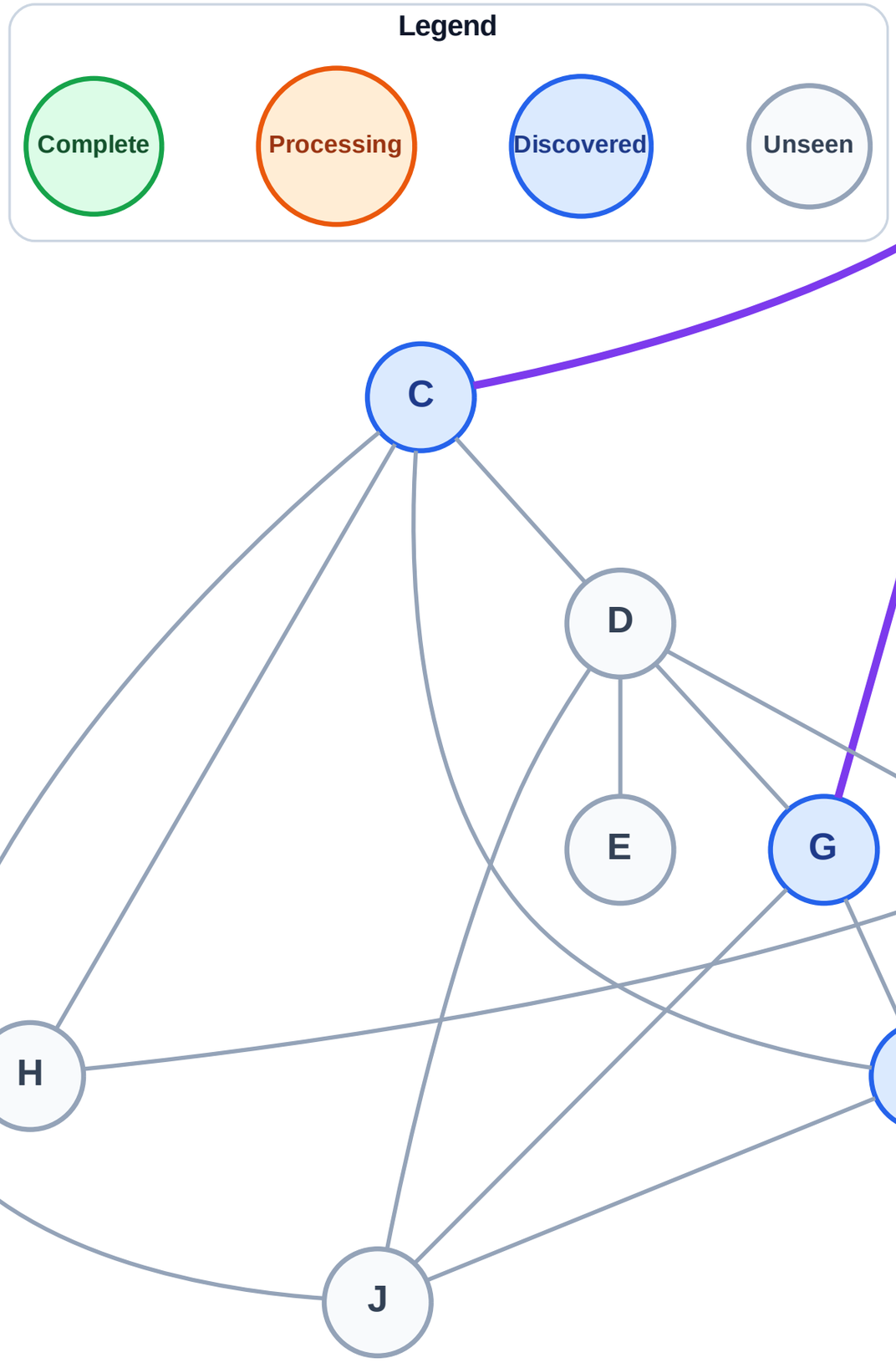


Stack (top at right): I | G | F | C | B

Visited: A

DFS Traversal

Step 4: Process node B

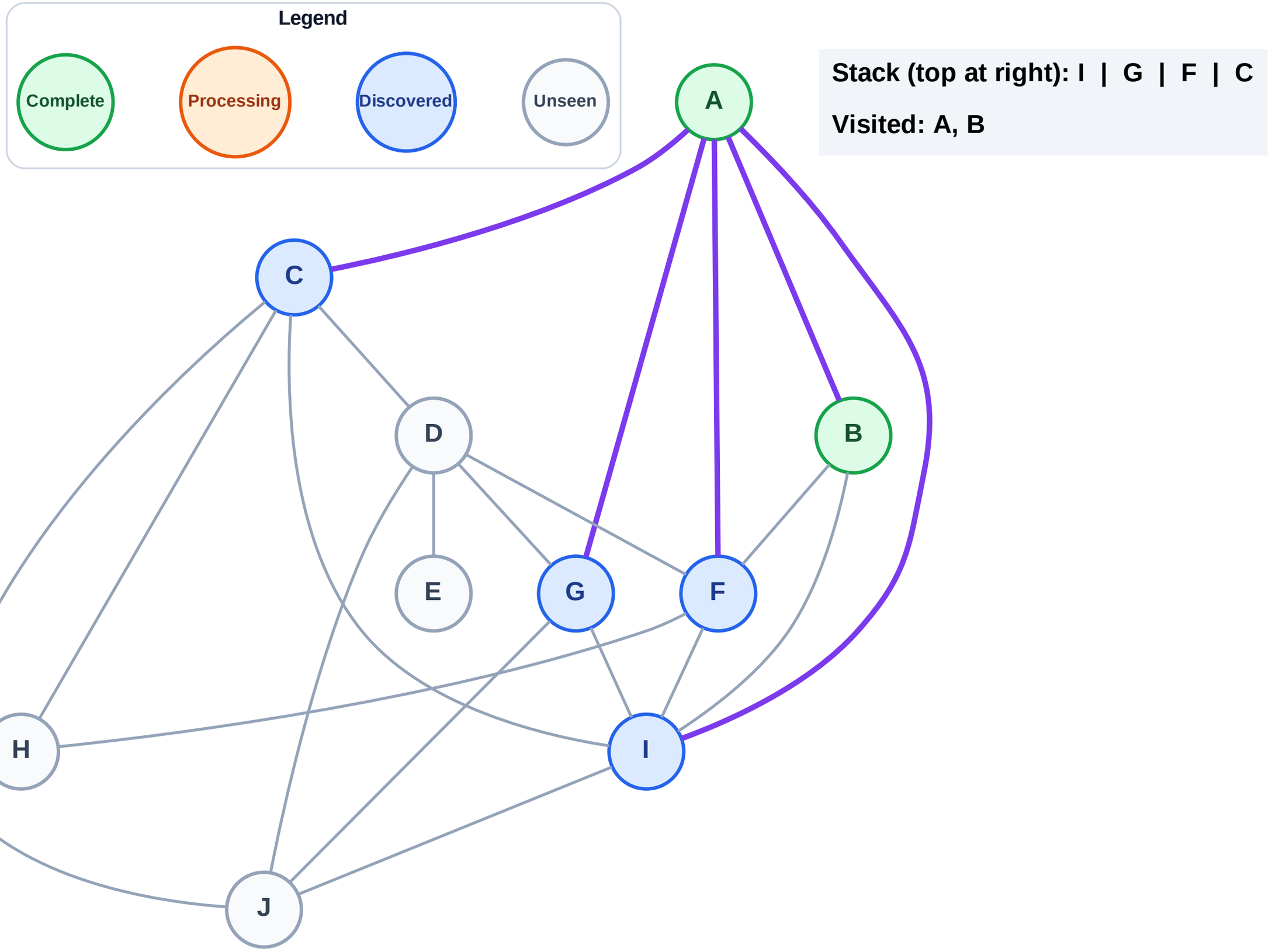


Stack (top at right): I | G | F | C

Visited: A

DFS Traversal

Step 5: No new neighbors from B

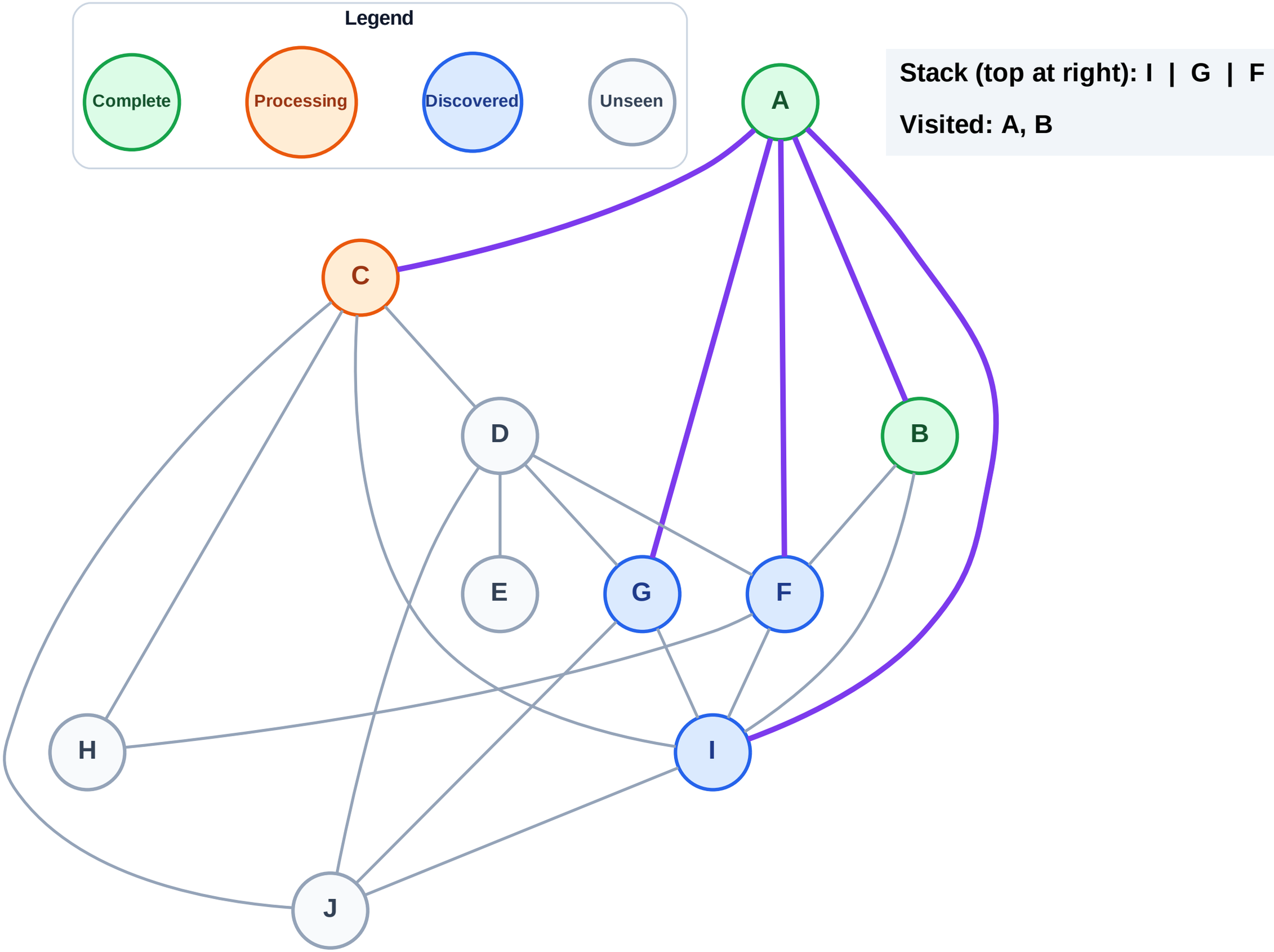


Stack (top at right): I | G | F | C

Visited: A, B

DFS Traversal

Step 6: Process node C

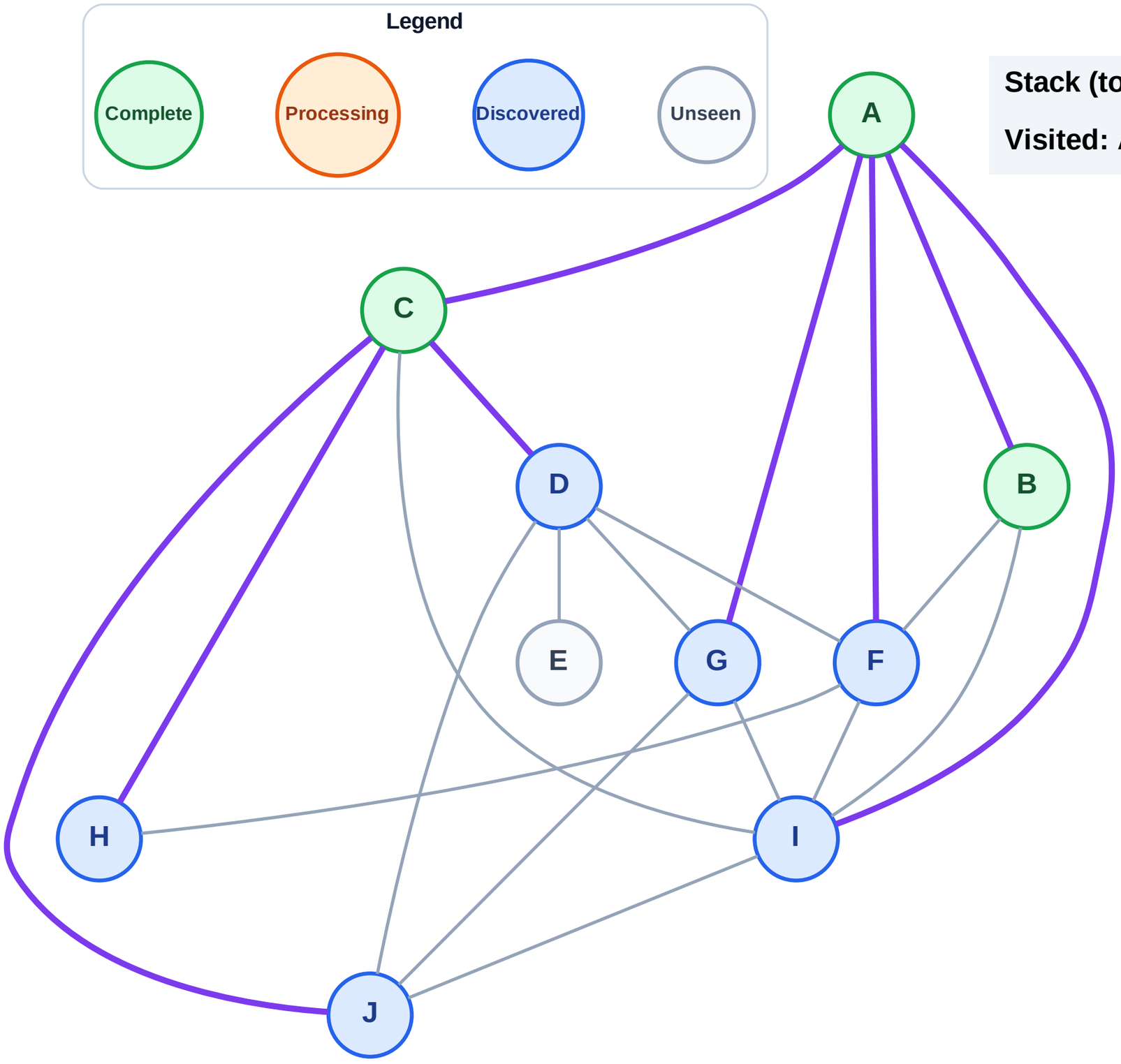


Stack (top at right): I | G | F

Visited: A, B

DFS Traversal

Step 7: Discover J, H, D from C

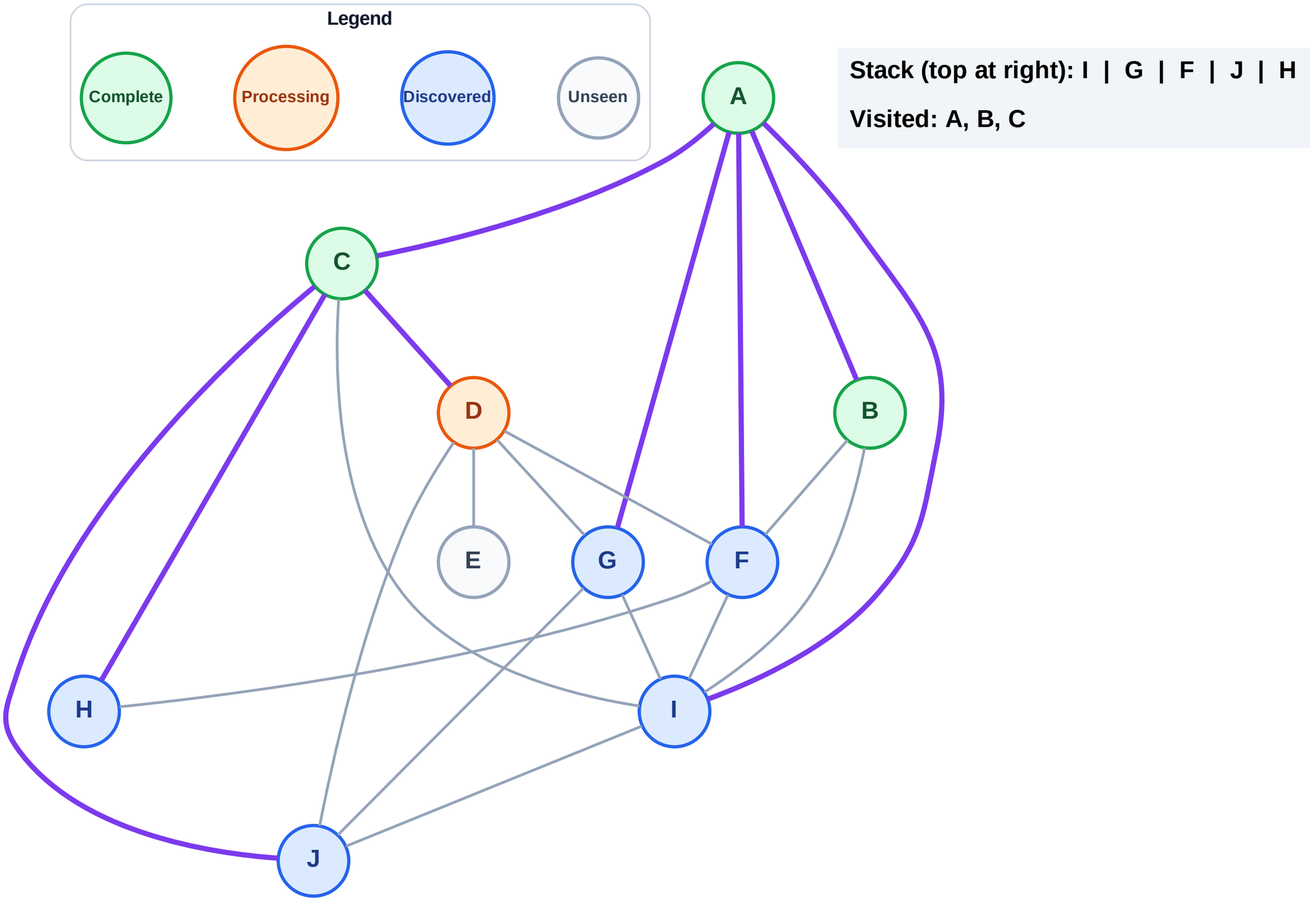


Stack (top at right): I | G | F | J | H | D

Visited: A, B, C

DFS Traversal

Step 8: Process node D

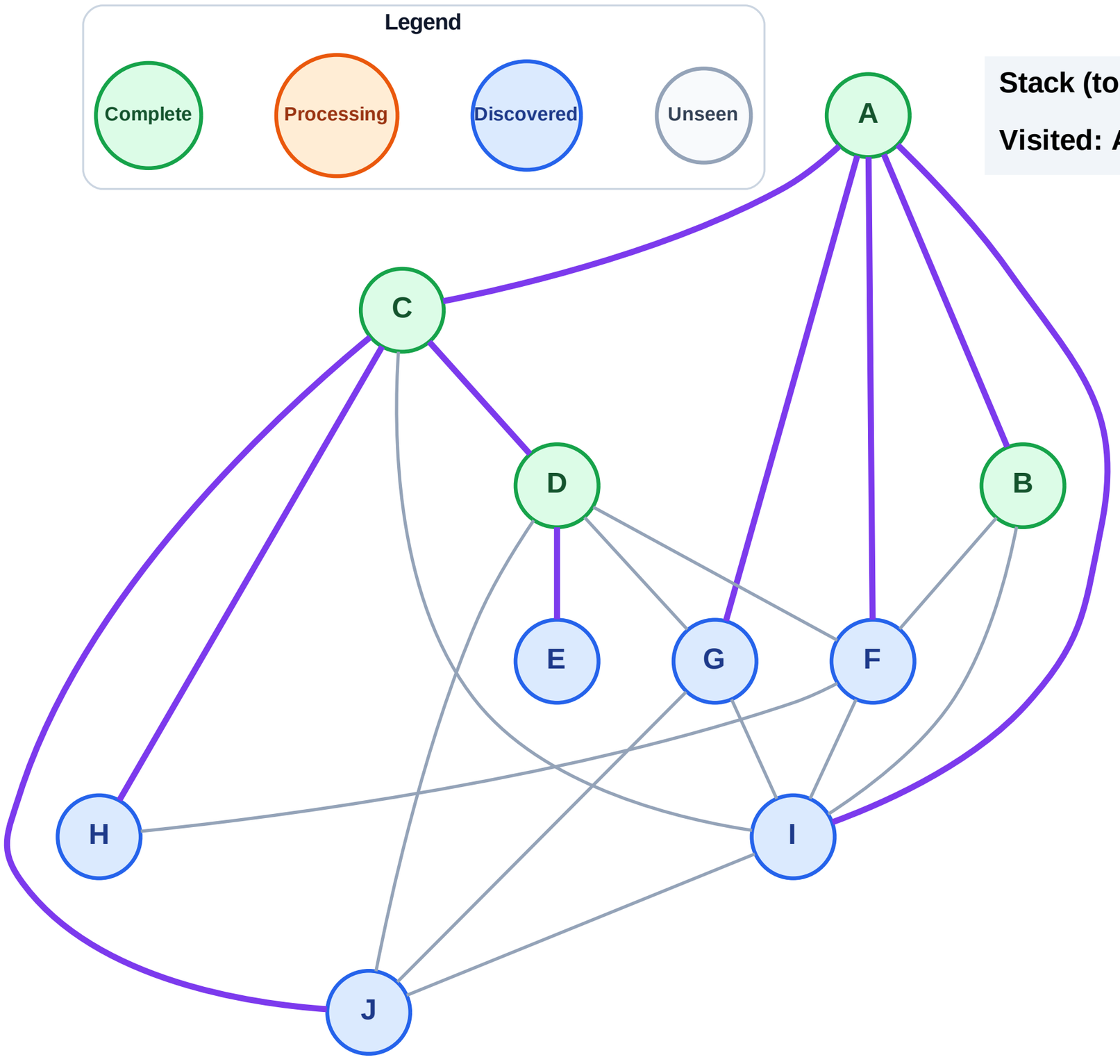


Stack (top at right): I | G | F | J | H

Visited: A, B, C

DFS Traversal

Step 9: Discover E from D

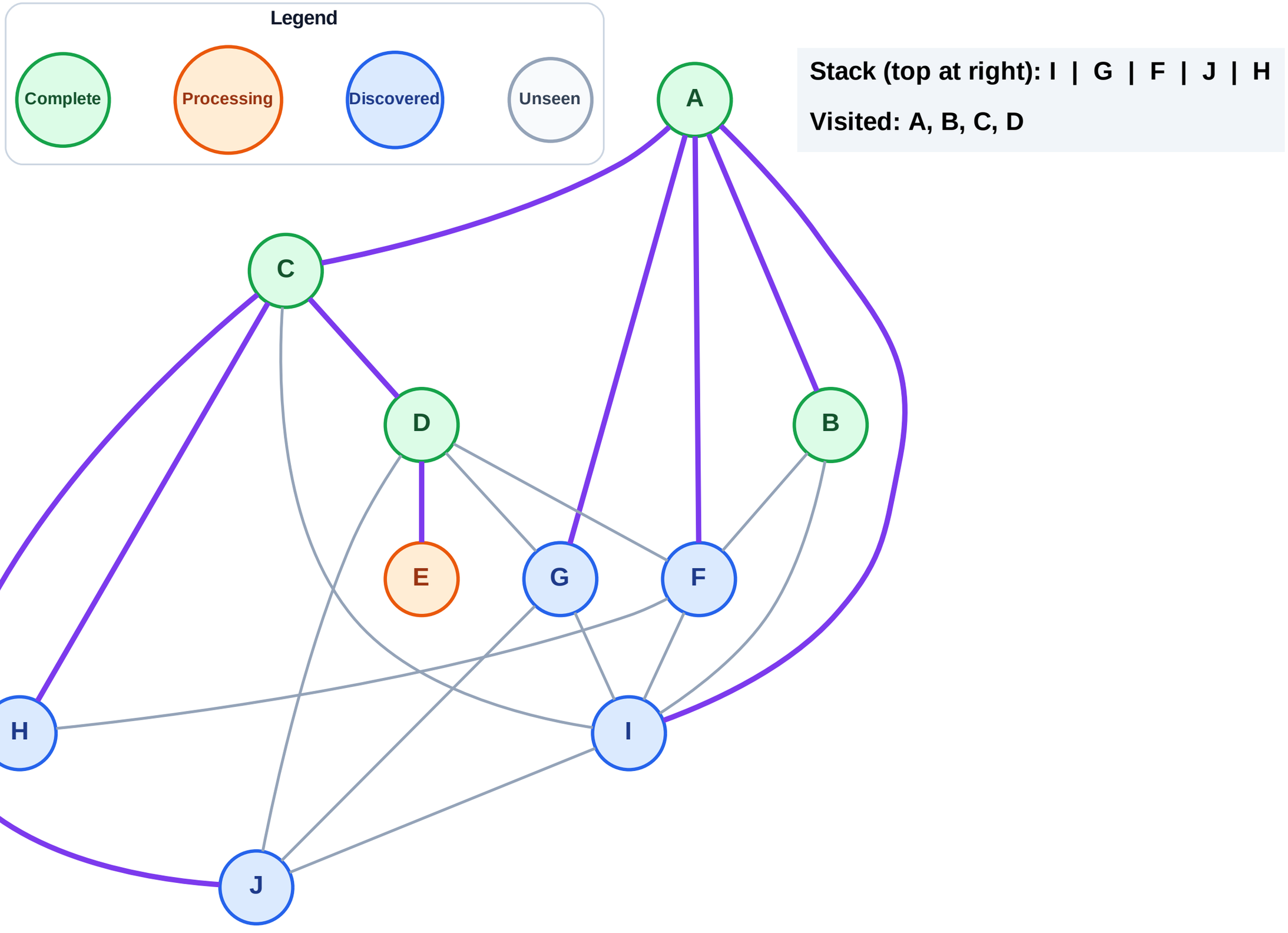


Stack (top at right): I | G | F | J | H | E

Visited: A, B, C, D

DFS Traversal

Step 10: Process node E

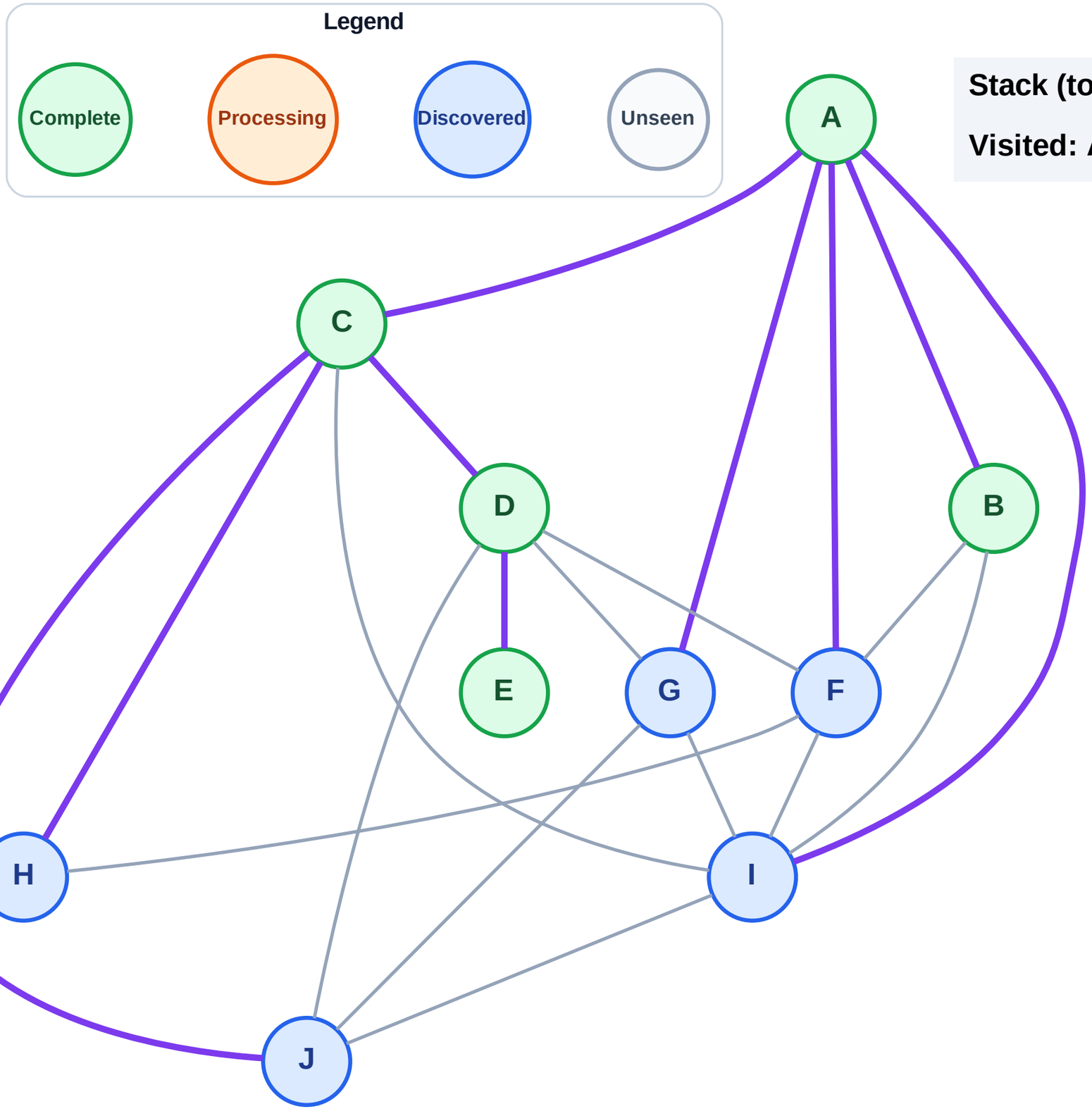


Stack (top at right): I | G | F | J | H

Visited: A, B, C, D

DFS Traversal

Step 11: No new neighbors from E

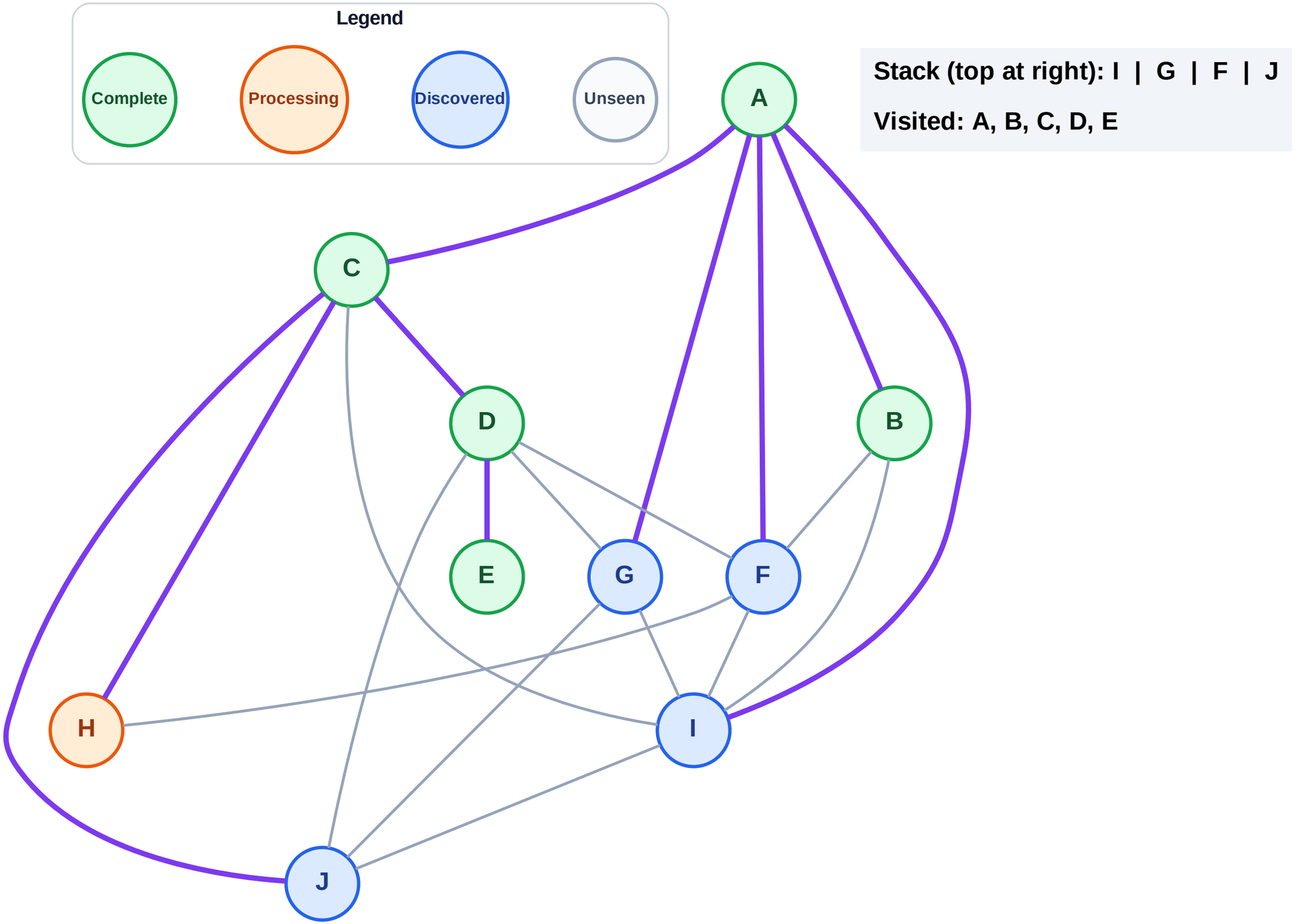


Stack (top at right): I | G | F | J | H

Visited: A, B, C, D, E

DFS Traversal

Step 12: Process node H

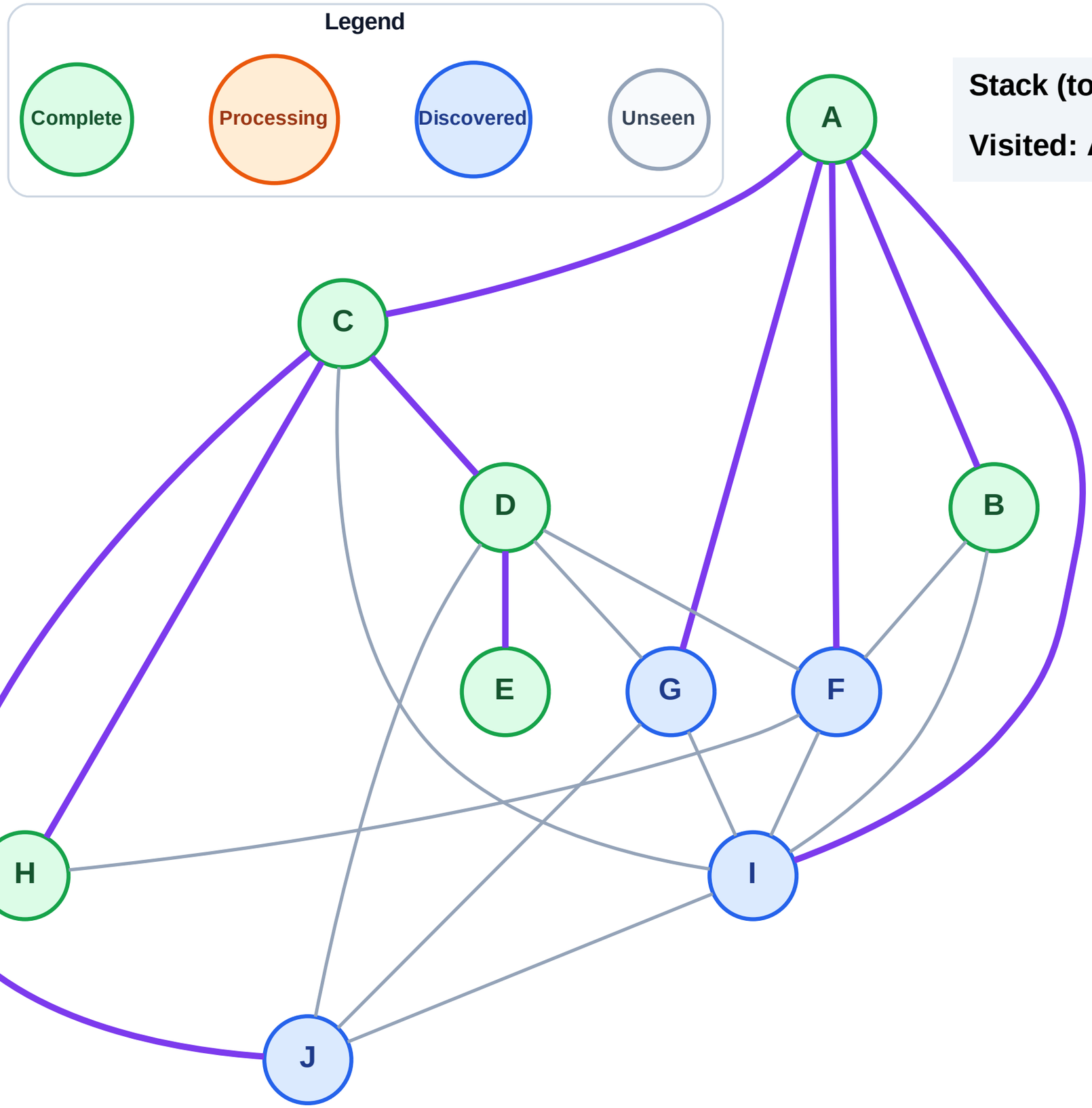


Stack (top at right): I | G | F | J

Visited: A, B, C, D, E

DFS Traversal

Step 13: No new neighbors from H

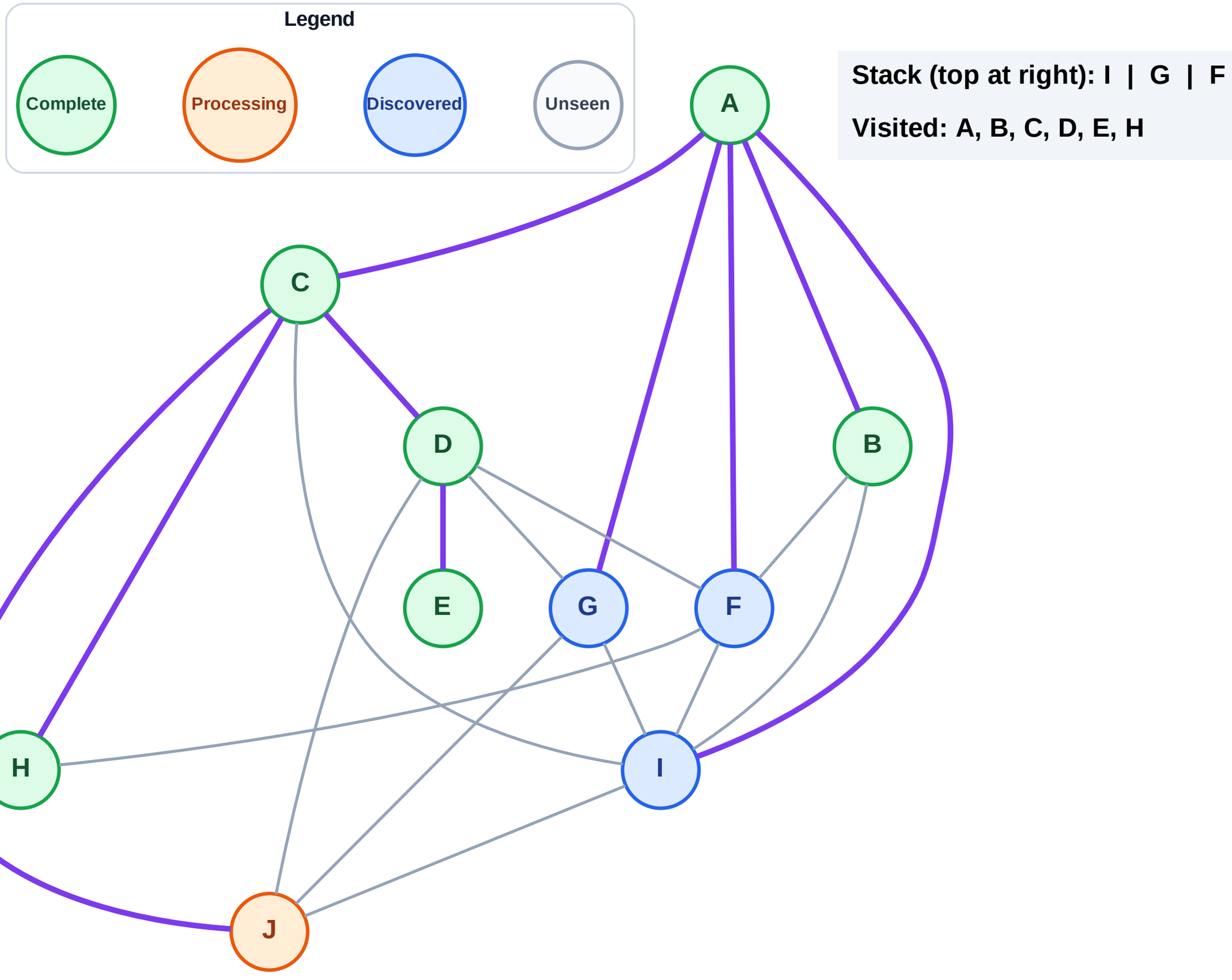


Stack (top at right): I | G | F | J

Visited: A, B, C, D, E, H

DFS Traversal

Step 14: Process node J

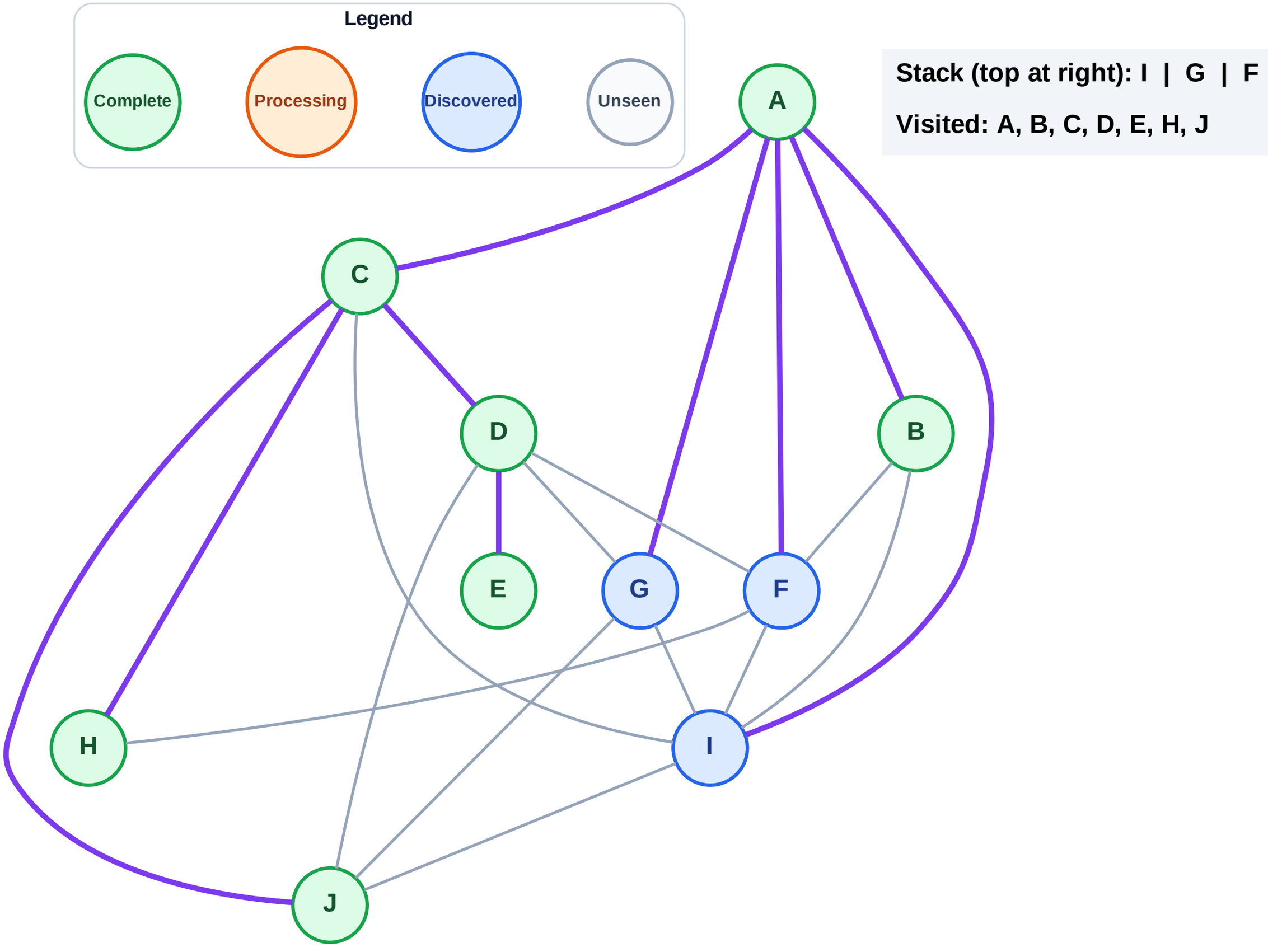


Stack (top at right): I | G | F

Visited: A, B, C, D, E, H

DFS Traversal

Step 15: No new neighbors from J

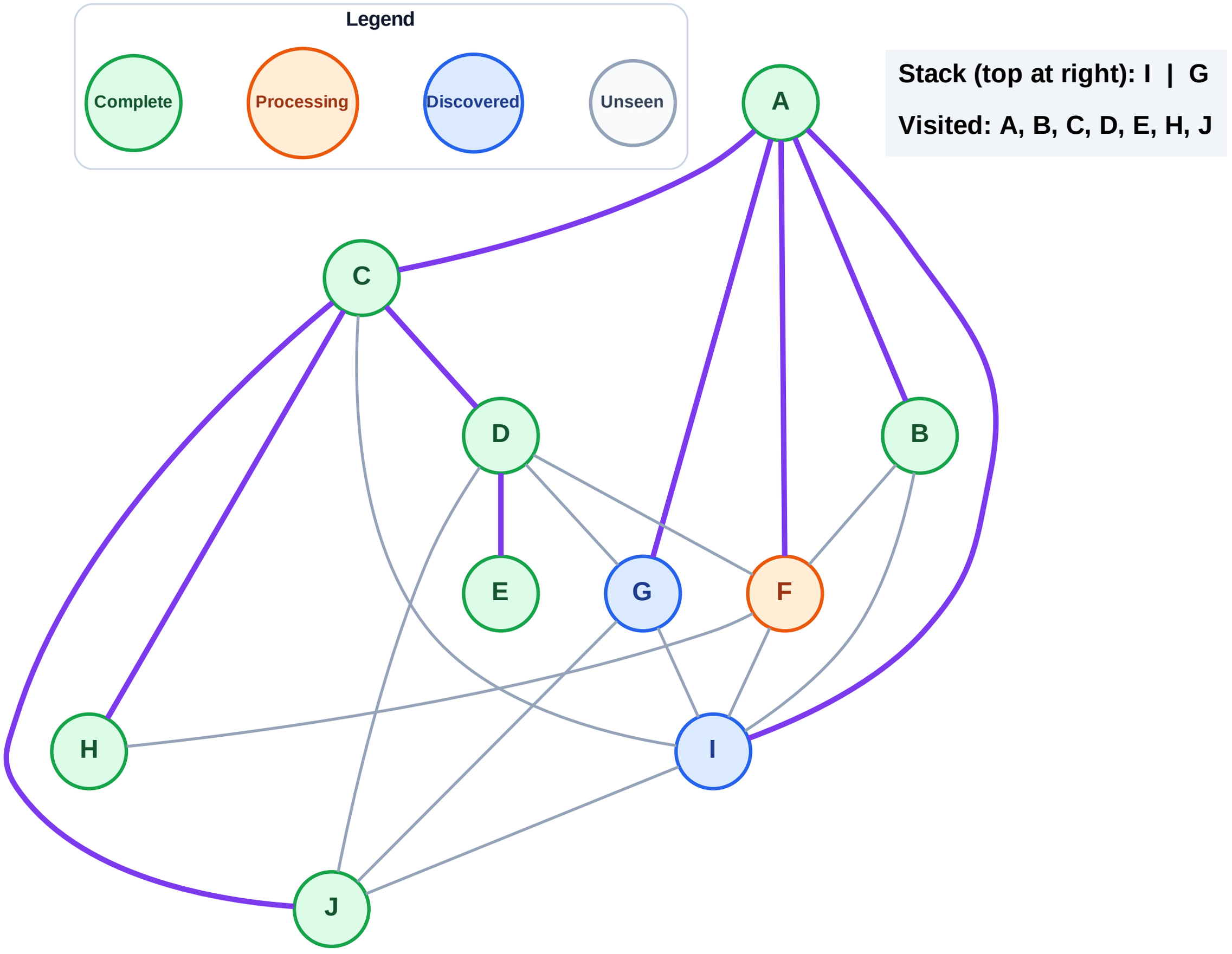


Stack (top at right): I | G | F

Visited: A, B, C, D, E, H, J

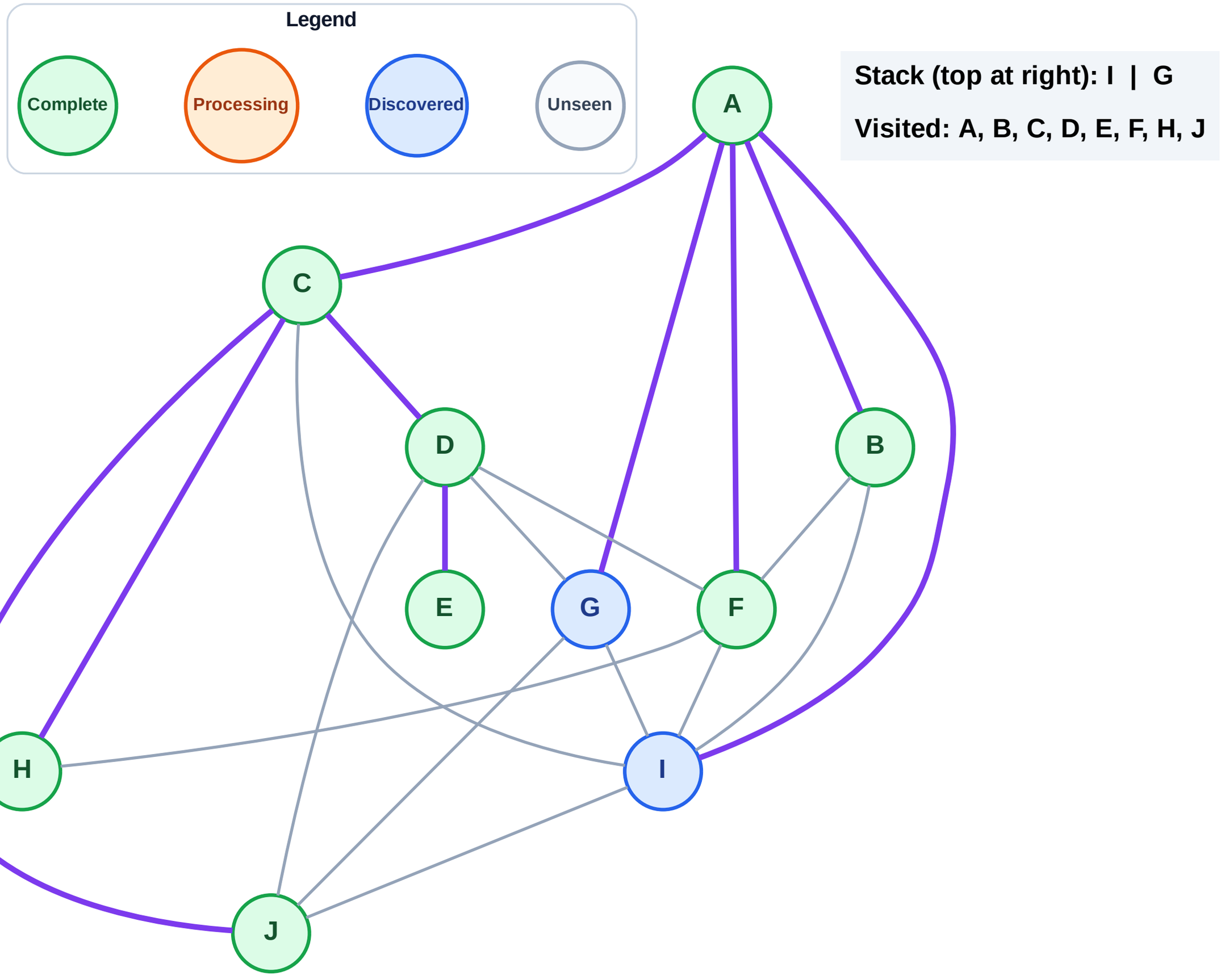
DFS Traversal

Step 16: Process node F



DFS Traversal

Step 17: No new neighbors from F

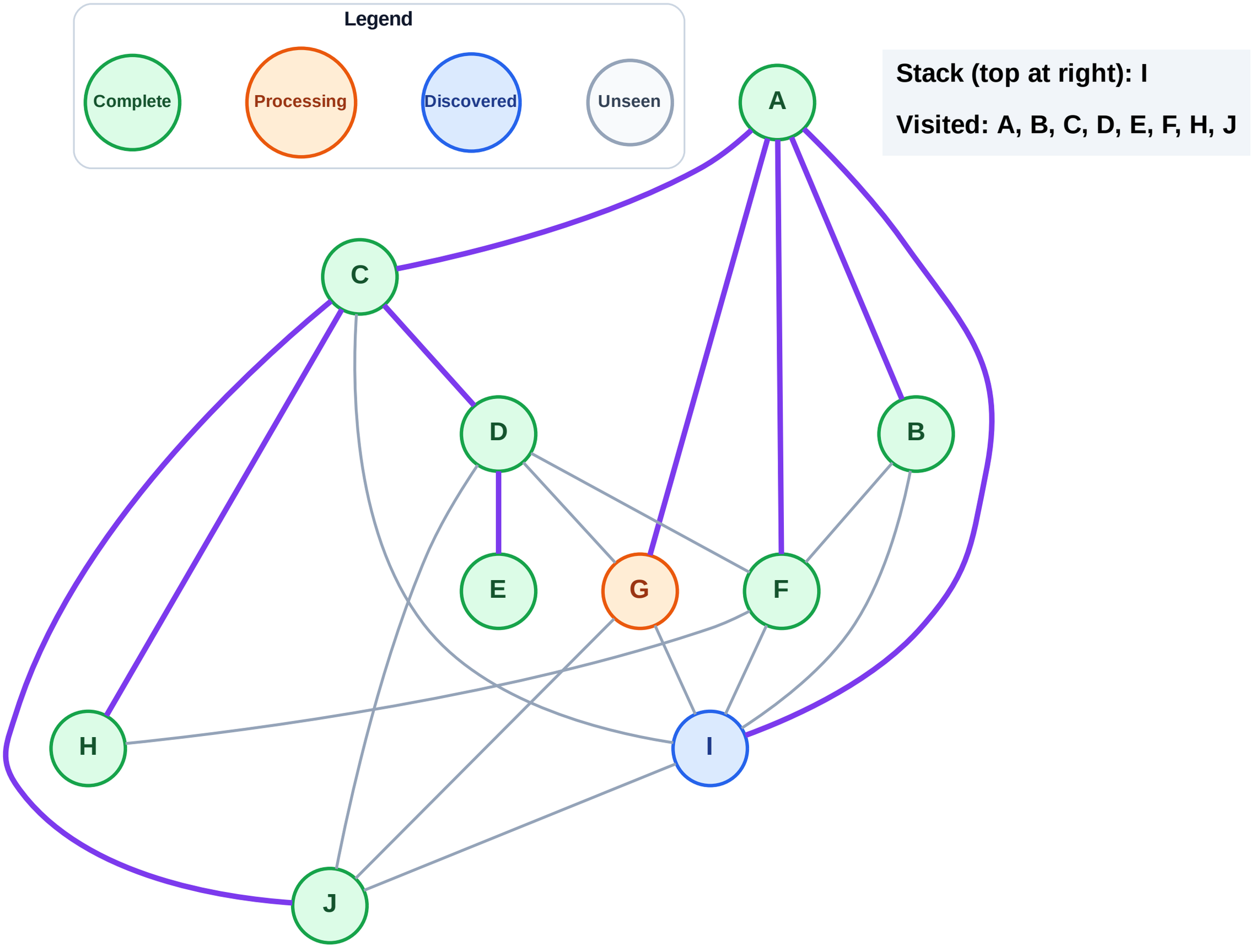


Stack (top at right): I | G

Visited: A, B, C, D, E, F, H, J

DFS Traversal

Step 18: Process node G

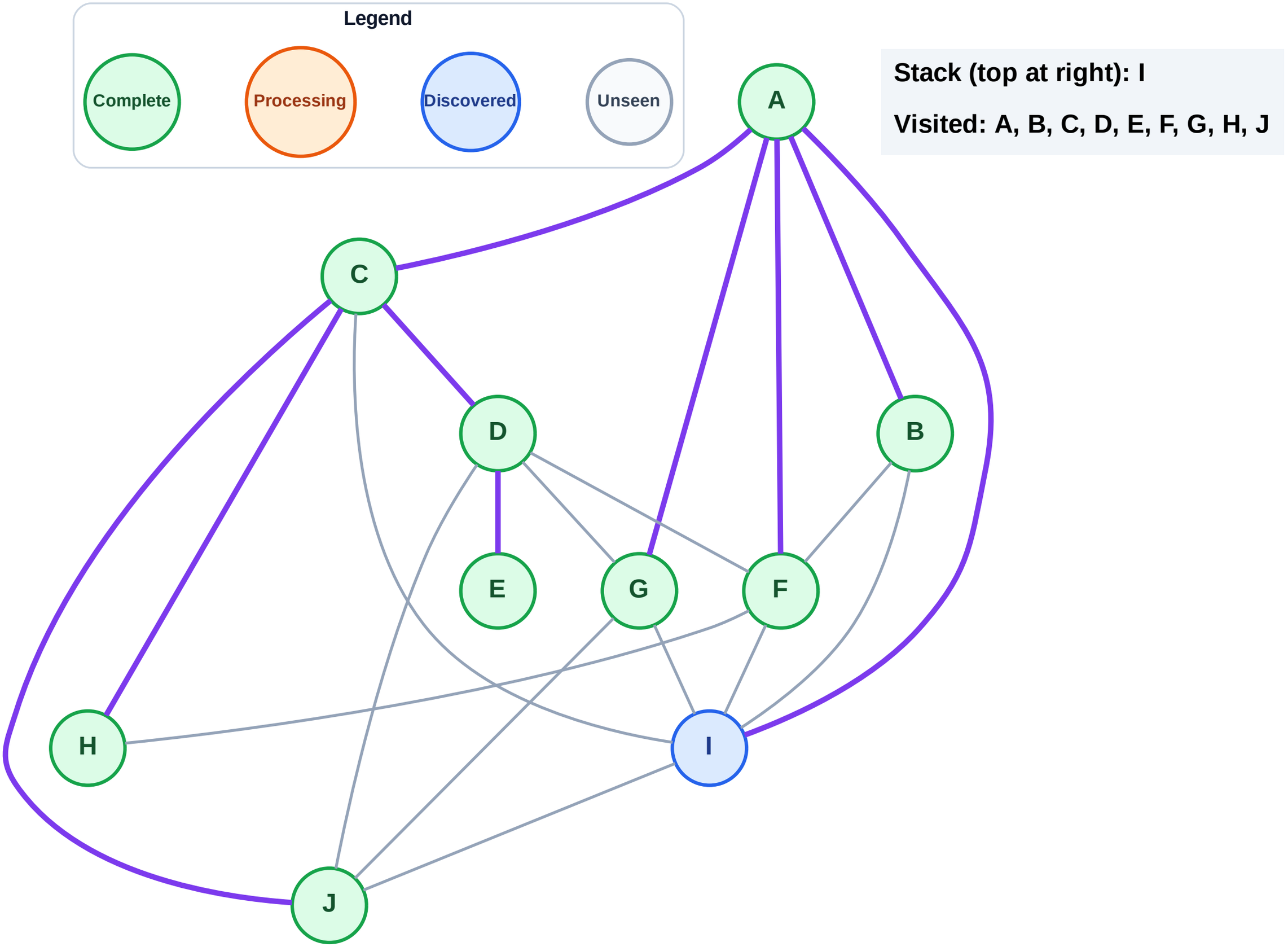


Stack (top at right): I

Visited: A, B, C, D, E, F, H, J

DFS Traversal

Step 19: No new neighbors from G

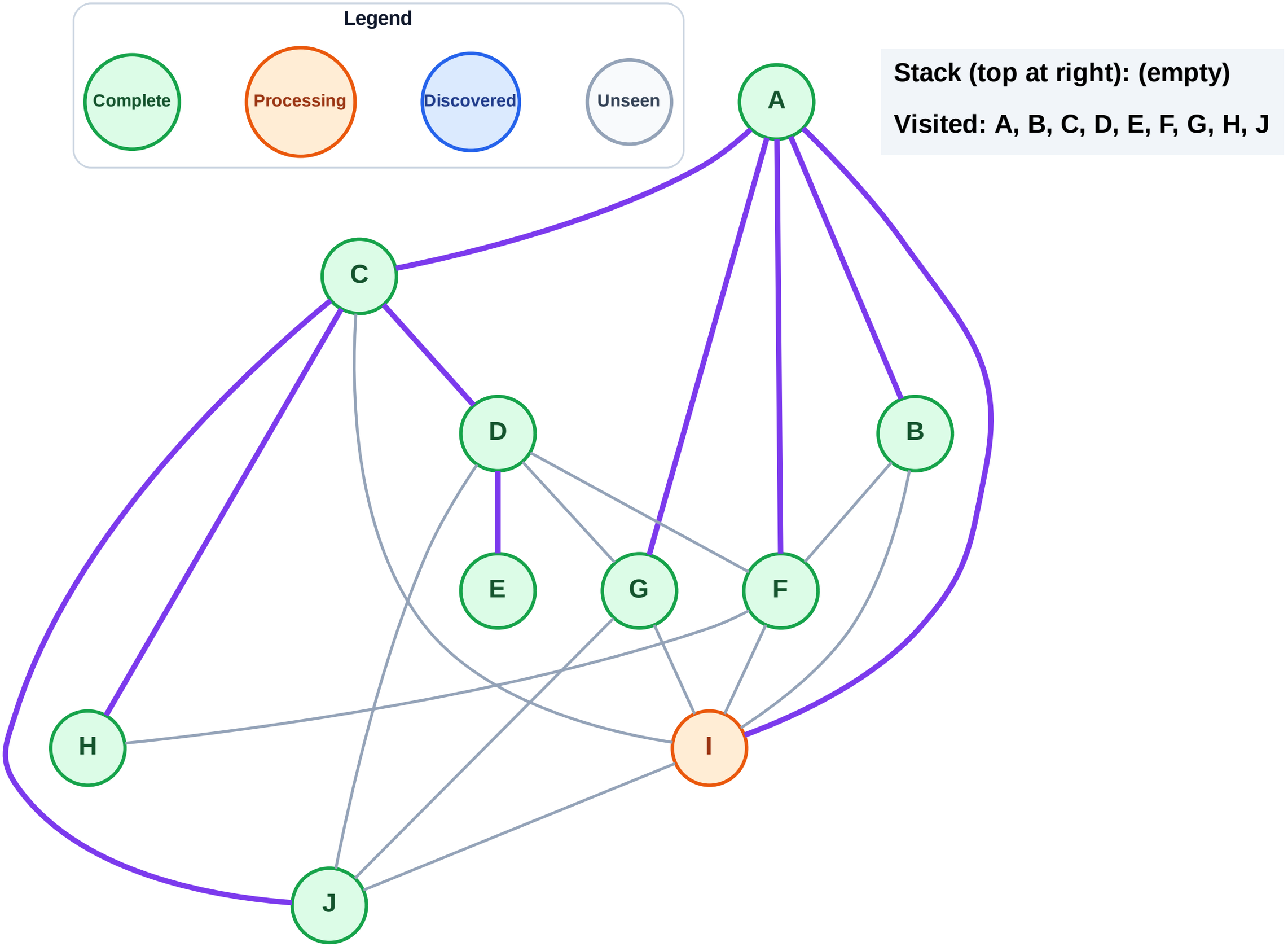


Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J

DFS Traversal

Step 20: Process node I

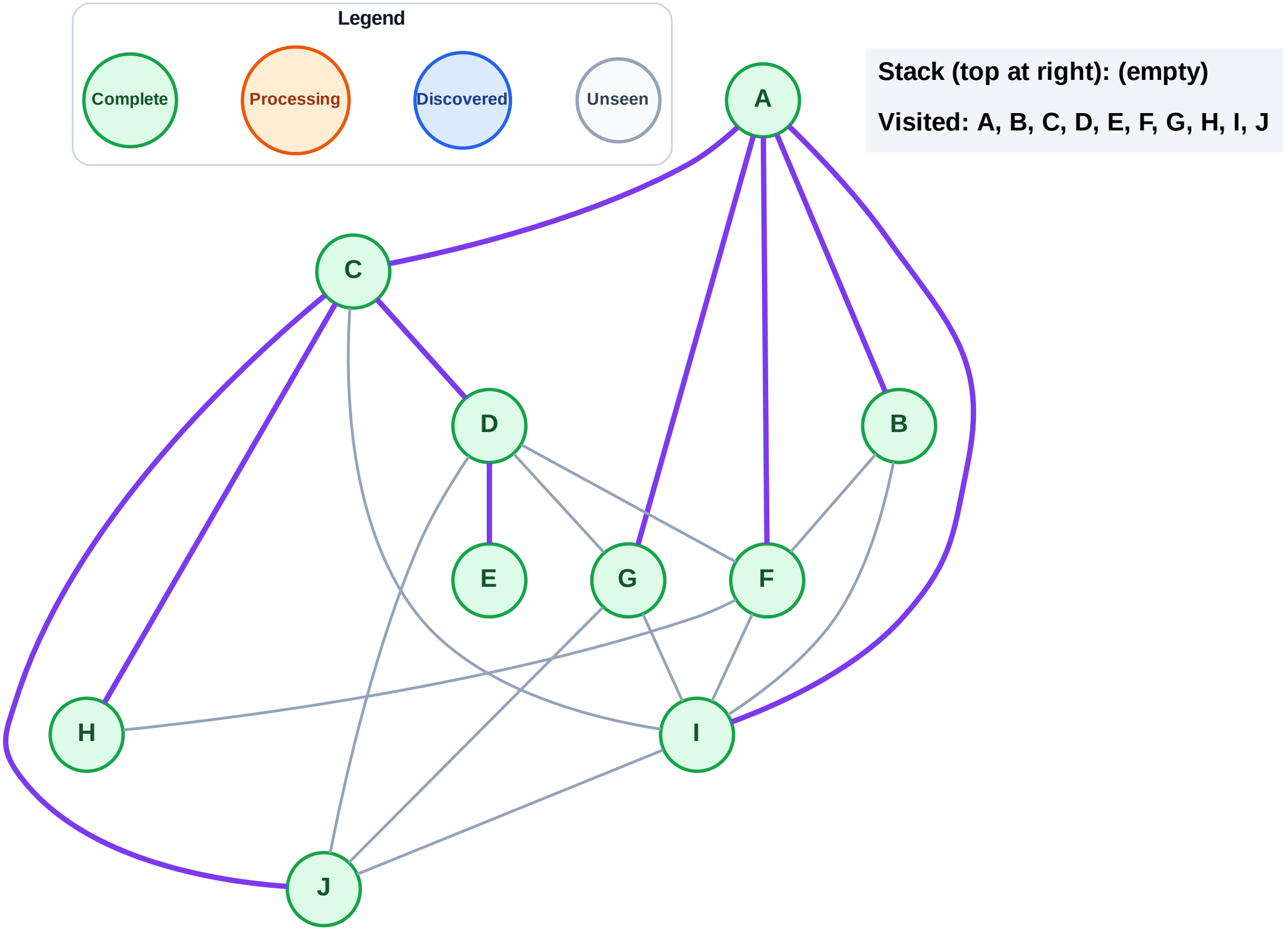


Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J

DFS Traversal

Step 21: No new neighbors from I



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J